APP-BASED COMMUNICATION AND EMOTIONAL JOURNEYS IN HEALTH AND EXERCISE APPLICATIONS: A STIMULATED RECALL STUDY

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ABSTRACT

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Abstract

This thesis examines how app-based communication shapes users' emotional journeys in health and exercise applications. As digital health and fitness services become increasingly embedded in everyday life, users encounter communication not only through explicit messages, notifications, and feedback, but also through implicit cues such as format, silence, data presentation, and app logic. However, less is known about how these communicative elements shape users' emotions, motivation, engagement, and willingness to continue app use.

The study adopts a qualitative and interpretive research design. The empirical data consist of six semi-structured stimulated recall interviews. The interview design was informed by the Critical Incident Technique and supported by Plutchik's Wheel of Emotions as an emotion-reflection tool. The data were analysed using theoretical, latent, and reflexive thematic analysis.

The findings show that users interpreted app-based communication as a broad interactional process rather than as explicit message content alone. Communication supported engagement when it helped users feel recognised, guided, informed, and in control of their own app experience. By contrast, communication weakened the user–app relationship when it appeared unclear, poorly timed, overly pressuring, or insufficiently responsive to the user's situation. These findings suggest that emotionally engaging app journeys depend not only on what applications communicate, but also on how appropriately they respond to users' emotional, bodily, and contextual needs.

The thesis contributes to corporate communication research by conceptualising health and exercise applications as mediated organisational touchpoints through which users may perceive support, pressure, recognition, neglect, trust, or emotional distance. It concludes that emotionally intelligent app communication requires sensitivity to context, timing, bodily experience, transparency, restraint, and user agency. The findings offer insights for designing health and exercise app journeys that are engaging, trustworthy, and ethically responsible.

Key words
corporate communication, health applications, exercise applications, emotions, customer journey, communication, emotional intelligence, users, Critical Incident Technique, thematic analysis

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Tiivistelmä
Tämä pro gradu -tutkielma tarkastelee, miten sovellusten viestintä muovaa käyttäjien emotionaalisia käyttäjäpolkuja terveys- ja liikuntasovelluksissa. Digitaalisten terveys- ja liikuntapalveluiden yleistyessä arjessa käyttäjät kohtaavat viestintää paitsi viestien, ilmoitusten ja palautteen myös rakenteen, hiljaisuuden, datan esitystavan ja toimintalogiikan kautta. Vähemmän tiedetään siitä, miten nämä elementit vaikuttavat käyttäjien tunteisiin, motivaatioon, sitoutumiseen ja haluun jatkaa sovelluksen käyttöä.

Tutkimuksessa hyödynnettiin laadullista ja tulkinnallista tutkimusasetelmaa. Aineisto koostuu kuudesta puolistrukturoidusta virikehaastattelusta, joita ohjasivat kriittisten tapahtumien tekniikka ja Plutchikin tunnepyörä. Aineisto analysoitiin teoreettisen, latentin ja refleksiivisen temaattisen analyysin avulla.

Tulokset osoittavat, että käyttäjät tulkitsivat sovellusten viestinnän laajana vuorovaikutuksellisena prosessina, eivät vain eksplisiittisenä viestisisältönä. Viestintä tuki sitoutumista, kun se auttoi käyttäjiä tuntemaan itsensä huomioiduiksi, ohjatuiksi, informoiduiksi ja oman sovelluskokemuksensa hallitsijoiksi. Sen sijaan viestintä heikensi käyttäjän ja sovelluksen suhdetta, kun se oli epäselvää, huonosti ajoitettua, liian painostavaa tai käyttäjän tilanteeseen sopimatonta. Tulokset viittaavat siihen, että sitouttavat käyttäjäpolut riippuvat sekä siitä, mitä sovellukset viestivät, että siitä, miten tarkoituksenmukaisesti ne vastaavat käyttäjien emotionaalisiin, kehollisiin ja kontekstuaalisiin tarpeisiin.

Tutkielma edistää yritysviestinnän tutkimusta tarkastelemalla terveys- ja liikuntasovelluksia välittyneinä organisatorisina kosketuspisteinä, joiden kautta käyttäjät voivat kokea tukea, painetta, huomioiduksi tulemista, laiminlyöntiä, luottamusta tai emotionaalista etäisyyttä. Tutkielman perusteella emotionaalisesti älykäs sovellusviestintä edellyttää herkkyyttä kontekstille, ajoitukselle, kehollliselle kokemukselle, läpinäkyvyydelle ja käyttäjän toimijuudelle. Tulokset tarjoavat näkökulmia sitouttavien, luotettavien ja eettisesti vastuullisten terveys- ja liikuntasovellusten käyttäjäpolkujen suunnitteluun.

Asiasanat
yhteisöviestintä, terveyssovellukset, liikuntasovellukset, tunteet, asiakaspolku, viestintä, tunneäly, käyttäjät, kriittisten tapahtumien tekniikka, temaattinen analyysi

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1 INTRODUCTION

Why do some digital experiences make us feel energised and motivated while others make us feel frustrated or quietly disengaged? As technology becomes increasingly embedded in our lives through apps and other digital services, so too grows the influence of our emotional responses to interactions with these services. These emotional responses can shape how users remember interactions, evaluate trust, and form loyalty-related perceptions. Every tap, swipe, and notification involves a complex interplay of design, communication, and human emotion. Understanding the patterns of this interplay can inform more empathetic, engaging, and effective communication strategies.

As digital services proliferate and competition for user attention intensifies, understanding the customer journey has become essential for organisations aiming to support meaningful and sustained user relationships. While often conceptualised as a sequence of functional touchpoints—such as onboarding, usage, and customer service—the customer journey also contains a dynamic emotional layer. The emotional journey refers to the changing emotional states that emerge throughout users' interactions with a brand or service—states that may shape users' evaluations, motivation, trust, and loyalty-related perceptions. Studying this emotional dimension is crucial for developing communication strategies that are emotionally intelligent and experience driven. The emotional journey can be seen as an affective layer within the broader customer journey—one that is shaped not only by users' actions and interface design, but also by communicative cues such as tone, feedback, guidance and perceived responsiveness. Scholars such as Lemon and Verhoef (2016) have emphasised the importance of understanding customer journeys as multidimensional constructs that integrate emotional, cognitive, and behavioural experiences.

Existing research has mapped emotional responses across customer journeys using methods such as journey mapping, personas, and retrospective interviews (Rosenbaum et al., 2017; Mele et al., 2021). While valuable, these studies often prioritise service design or usability over communication. Følstad and Kvale (2018) note that the communicative content of touchpoints remains underexplored, despite its central role in shaping emotional responses. This lack of attention to in-the-moment messaging, tone, and feedback leaves a gap in

understanding how digital services influence emotional experiences through communication.

This thesis therefore does not evaluate health and exercise applications primarily as technical interfaces or app design products. Instead, it examines them as communicative environments in which organisational meaning is conveyed through messages, feedback, timing, silence, structure, and perceived responsiveness. From this perspective, features often associated with UX or app design become relevant to corporate communication when users interpret them as signs of support, neglect, pressure, recognition, or care.

This perspective is especially relevant in the context of health and exercise applications, which in this thesis refer to applications used in relation to exercise, training, recovery, bodily self-monitoring, or exercise-related reflection. These include, for example, activity trackers, recovery platforms, workout logging tools, and coaching applications, which are increasingly integrated into users' everyday exercise and self-monitoring practices. These applications offer users a variety of features such as activity tracking, personalised plans, gamification elements, and behavioural notifications (Esmaeilzadeh, 2021).

Despite their widespread use and commercial success, earlier studies have often overlooked how communication strategies within these applications contribute to emotional responses, perceived engagement, and users' willingness to continue app use. Addressing this gap is particularly relevant within the field of corporate communication, where understanding user experiences can help inform strategies that may support stronger relationships between users and organisations (Lemon & Verhoef, 2016).

1.1 Research aim and questions

This thesis investigates how communication shapes users' emotional experiences during interactions with health and exercise applications. The customer journey serves as the overarching theoretical framework, with particular attention paid to the communicative elements that influence users' emotional experiences. By employing Robert Plutchik's (1980) Wheel of Emotions, the study provides a structured tool for categorising emotional responses at key stages of app interaction, including, for example, opening the app, selecting a workout, completing a session, and receiving feedback or notifications. These moments align with critical incidents in the customer journey (Edvardsson & Roos, 2001, p. 253; Watkins et al., 2022, p. 710), where emotions such as anticipation, joy, frustration, or disappointment may shape how users interpret the app, evaluate the interaction, and describe their motivation or engagement.

This thesis approaches the emotional journey through the lens of corporate communication. Emotions are therefore not treated as byproducts of app design or functionality. The focus lies on how communicative choices—such as messages, feedback loops, notifications, tone, timing, silence, guidance, and perceived responsiveness—shape the emotional landscape of the user experience. In this study, app-based touchpoints are therefore examined as communicative acts

through which users may perceive support, pressure, recognition, neglect, trust, or emotional distance.

This boundary is important for positioning the study within corporate communication. The thesis does not ask whether the applications are technically well designed, nor does it seek to measure users' emotions as psychological states. Instead, it examines communication as a strategic, relational, and emotionally consequential process within digital service encounters. By identifying how users interpret emotionally significant touchpoints, the study contributes to understanding how more empathetic, context-sensitive, and emotionally intelligent communication strategies can be developed for health and exercise applications.

The study is guided by the broader research problem of understanding how communication can shape and support emotionally engaging user journeys in health and exercise applications. This problem is examined through three interrelated research questions:

RQ1: What emotional responses do users report at different stages of interaction with health and exercise applications, and how are these shaped by app-based communication?

RQ2: How do users perceive the relationship between app-based communication, emotional responses, motivation, and engagement?

RQ3: How can users' emotional journeys inform emotionally intelligent communication strategies in health and exercise applications?

To answer these questions, this study adopts a qualitative approach, with data collected through stimulated recall interviews supported by Plutchik's (1980) Wheel of Emotions. Participants were invited to revisit recent health or exercise app experiences and identify emotionally significant moments that occurred before, during, or after app use. These moments were examined as critical incidents through which communicative elements, such as feedback, notifications, guidance, tone, timing, format, silence, and personalisation, can be analysed in relation to users' reported emotional responses. The goal is to connect insights from user experience research with strategic communication by showing how app-based touchpoints can function as emotionally consequential communication within digital service environments.

The thesis proceeds from theoretical framing to methodology, findings, and discussion. It first introduces the key concepts used to analyse emotional journeys in health and exercise applications, then presents the qualitative research design and thematic analysis. The findings are reported through six themes and are followed by a discussion of theoretical contributions, practical implications, study evaluation, future research, and final conclusions. The use of generative AI and AI-enhanced tools during the thesis process is described transparently in Appendix 1.

2 LITERATURE REVIEW

Human experiences are inherently ambiguous, fleeting, and emotionally charged. As Palmer (2010) argues, customer experiences are not stable or predictable constructs, but hedonistic and novel phenomena — their value dictated by the individual undergoing them and shaped through communicative interactions. This perspective challenges customer experience models that emphasise structured stages, rational progression, or relative emotional consistency (Gahler et al., 2022; Voorhees et al., 2017). Instead, it invites us to consider the transitory nature of emotions as central to how people engage with digital services and communication touchpoints.

This view forms the philosophical foundation of this thesis. If each emotional response is shaped by a moment's context, then every communicative act — every notification, prompt, or piece of guidance — becomes a potentially transformative experience. Therefore, strategic communication is not merely a functional layer in service delivery; it is part of how users interpret, evaluate, and emotionally respond to digital service encounters. This is particularly relevant for the field of corporate communication, where meaning is not fixed but co-constructed in context-sensitive interactions.

This broad understanding of communication is supported by Watzlawick et al.'s (1967) interactional view of communication, particularly their first axiom that "one cannot not communicate". From this perspective, communication is not limited to intentional verbal messages, but includes behaviours, absences, timing, responses, and contextual cues that may be interpreted as meaningful within an interaction. Applied to health and exercise applications, this means that communicative meaning may emerge not only from explicit messages such as notifications or feedback, but also from silence, delay, generic responses, interface prompts, or the perceived lack of recognition. This perspective supports the thesis's treatment of app-based touchpoints as communicative encounters that can shape users' emotional interpretations.

The theoretical framework of this thesis is organised hierarchically. The customer journey and emotional journey form the primary framework, as they locate users' emotional experiences across app-based touchpoints and explain why

these experiences matter for user–organisation relationships. App-based communication provides the corporate communication lens through which these touchpoints are interpreted. Emotionally intelligent communication, emotional granularity, and the theory of constructed emotion function as interpretive concepts that help explain how users make sense of emotional and communicative cues. By contrast, Plutchik's Wheel of Emotions and the Critical Incident Technique are used primarily as methodological and analytical tools: Plutchik's model supports the naming and categorisation of reported emotions, while CIT helps identify emotionally significant interaction moments. This hierarchy guides the structure of the theory chapter and clarifies how the different concepts contribute to the empirical analysis.

2.1 Contextualising health and exercise applications as communicative environments

Health and exercise applications have become central to many users' routines, offering features such as activity tracking, gamification, personalised plans, recovery monitoring, workout logging, and behavioural notifications (Bitrián et al., 2020; Esmaeilzadeh, 2021; Jang et al., 2018). In line with the scope of this thesis, these applications are understood broadly as apps used in relation to exercise, training, recovery, bodily self-monitoring, or exercise-related reflection. Although such features are often designed to support motivation and continued use, many users abandon health and exercise applications within weeks or months (Esmaeilzadeh, 2021; Feng et al., 2020). This suggests that app engagement cannot be understood only through functionality, but also through the emotional and communicative quality of the user experience.

Existing research has shown that gamified features, such as leaderboards and badges, can support satisfaction and motivation in the short term (Bitrián et al., 2020). Notifications are also widely used to encourage continued engagement, yet their effectiveness varies depending on tone, timing, relevance, and user context (Feng et al., 2020; Yin et al., 2022). When poorly executed, such features may create emotional dissatisfaction and contribute to disengagement (Bougie et al., 2003). These findings point to a need to examine not only whether app features function, but also how users interpret them emotionally.

From a communication management perspective, Seiffert-Brockmann et al. (2018) similarly show that gamified mobile service applications can influence stakeholder engagement through users' psychological involvement and behavioural app use. Their study is useful for this thesis because it treats mobile apps as communication instruments and highlights that engagement depends on users' motivations as well as on the presence of gamified features.

While app design and functionality have been studied extensively, less attention has been paid to how communicative elements embedded within these applications shape users' emotional experiences. In this thesis, such elements include explicit messages and notifications, as well as feedback, silence, timing, format, guidance, and perceived responsiveness. From a corporate communication

perspective, these elements matter because they can influence whether users experience the application as supportive, pressuring, generic, trustworthy, or emotionally distant.

Recent corporate communication and public relations research support this broader view of digital communication. Brockhaus et al. (2022) argue that communication technology and digital infrastructure have become part of how corporate communication is organised and practised. Lock (2019) similarly argues that digital environments require renewed attention to communicative organisation–stakeholder relationships, because relationships in online settings are constituted through mediated communication. In this thesis, these perspectives support the interpretation of health and exercise applications as digital environments where organisational communication is encountered through repeated app-based interactions, data, feedback, automation, and absence. Wiencierz and Röttger (2017) further show that big data can support corporate communication by generating stakeholder insights that inform more effective communication strategies. This perspective is relevant because health and exercise applications communicate not only through messages, but also through data-based feedback, personalisation logic, and automated interpretations of user behaviour.

Understanding the emotional consequences of app-based communication is increasingly relevant as digital health and wellness services continue to expand (Kari et al., 2016). Since exercise applications are also used in relation to public health goals (Krebs & Duncan, 2015), emotionally engaging and ethically responsible communication may matter not only for individual app satisfaction, but also for how users experience support, motivation, and trust in digital health-related services. A deeper focus on users' emotional experiences can help organisations design digital interactions that may support perceived engagement, motivation, trust, and more sustainable user relationships.

This thesis addresses these gaps by examining the emotional journey as it unfolds during interactions with health and exercise applications. The study investigates how users describe app-based communicative elements as shaping reported emotional responses at key interaction points, and how these responses are perceived in relation to motivation, engagement, trust, and willingness to continue app use. In doing so, the thesis contributes to corporate communication by examining digital service interaction as an emotionally consequential form of organisational communication.

Figure 1 summarises the conceptual framework of the study and illustrates how the main theoretical and methodological concepts are positioned in relation to one another. The customer journey and emotional journey provide the primary framework for locating app-based interactions across time and touchpoints. This perspective draws primarily on customer journey and engagement research (Følstad & Kvale, 2018; Lemon & Verhoef, 2016; Mele et al., 2021). Emotionally intelligent communication forms part of the corporate communication lens for interpreting how those touchpoints may support, pressure, guide, validate, or distance users. This part of the framework is informed by emotional intelligence and organisational communication literature (Goleman, 1995, 1998; Hess & Bacigalupo, 2011; Mayer & Salovey, 1997; Sigmar et al., 2012). The constructionist understanding of emotion and emotional granularity explains why emotional

responses are treated as contextual and interpretive rather than fixed reactions (Barrett, 2017; Tan et al., 2022). Plutchik's (1980) Emotion Model is presented separately because it functions as a practical emotion-reflection and categorisation tool rather than as the main theory of emotion formation.

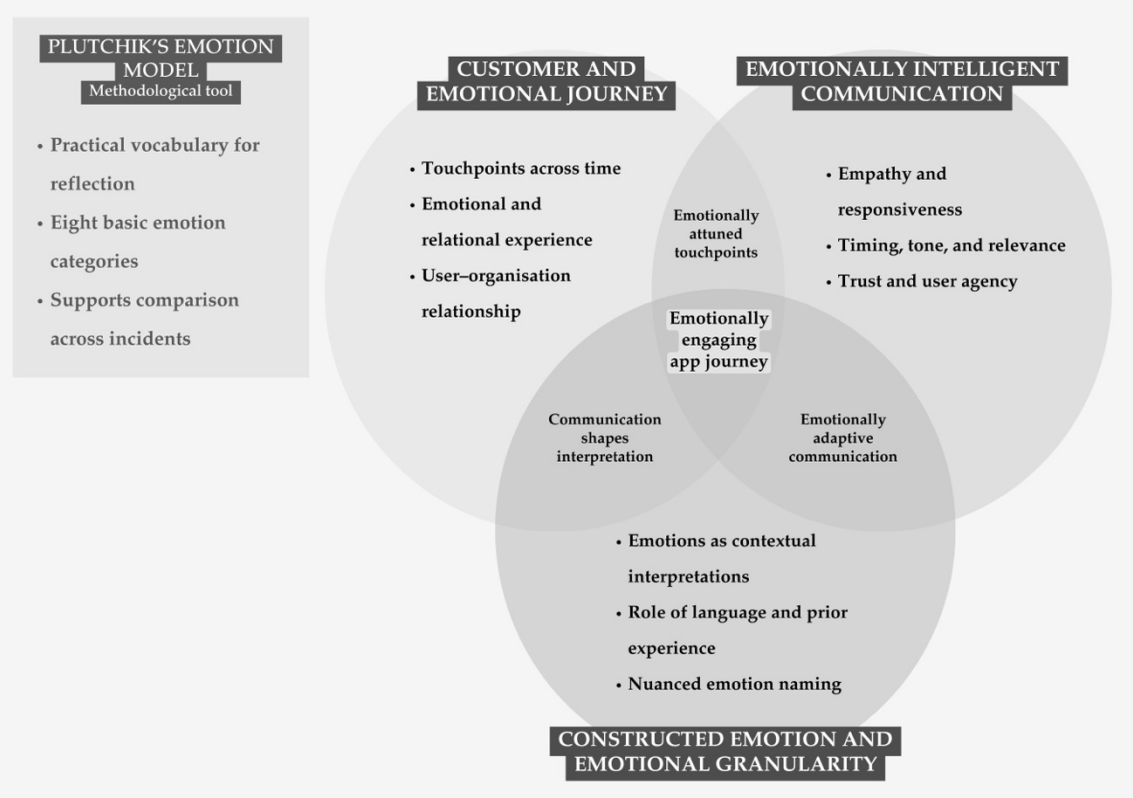


FIGURE 1. Conceptual framework of the study.

Figure 1 illustrates how the main concepts of the thesis overlap in the analysis of emotionally engaging app journeys. The customer journey and emotional journey perspectives define the central object of analysis by locating users' emotional experiences across app-based touchpoints. Emotionally intelligent communication and the broader understanding of app-based communication as mediated organisational communication provide the corporate communication lens for examining how these touchpoints may support, pressure, guide, validate, or distance users. The theory of constructed emotion and emotional granularity clarify why reported emotions are treated as contextual, interpreted, and linguistically mediated experiences. Plutchik's model is positioned separately because it functions as a practical tool for emotion reflection and categorisation rather than as the main theory of emotion formation. The following sections examine these concepts in turn, beginning with customer and emotional journeys in digital service encounters, before moving to emotionally intelligent communication, emotion reflection, emotional granularity, Plutchik's model, and the Critical Incident Technique.

2.2 Customer and emotional journeys in digital service encounters

Lemon and Verhoef (2016) conceptualise customer experience as a customer's journey with an organisation over time. This journey unfolds across multiple touchpoints and stages of interaction, and it is shaped not only by the immediate service encounter but also by past experiences, external influences, and later evaluations. Importantly, customer experience is multidimensional: it includes cognitive, emotional, behavioural, sensorial, and social responses to an organisation's offerings. In the context of health and exercise applications, the user journey is therefore not limited to the workout itself. It may also include opening the app, receiving reminders, interpreting progress feedback, reacting to achievement messages, reviewing recovery data, or returning to the app after a period of inactivity.

This broader understanding is useful because it moves attention away from single moments of app use. Instead, it makes it possible to examine how different touchpoints accumulate, interact, and become meaningful over time. Norton and Pine (2013) describe customer journey mapping as a strategic tool for innovation, as it can help organisations identify where resources and communication efforts should be directed. However, while journey maps may reveal common patterns across user segments, emotionally significant moments often become visible only when individual experiences are examined in more detail (Følstad & Kvale, 2018; Mele et al., 2021). This distinction is important for this thesis, which focuses on concrete app-related moments rather than general evaluations of health and exercise applications.

Følstad and Kvale (2018) further broaden customer journey thinking by emphasising both observable and experiential aspects of the journey, particularly in digital service contexts. Their perspective foregrounds the lived experience of the user and highlights how emotional and perceptual dimensions shape service satisfaction. Traditional customer journey models have often focused on functional stages or transactional milestones, such as awareness, consideration, purchase, and post-purchase evaluation (Voorhees et al., 2017). More recent research, however, has expanded this view to include emotional, relational, and communicative dimensions. Mele et al. (2021), for example, argue that customer engagement unfolds dynamically across touchpoints and may be shaped through subtle emotional and social cues.

In this thesis, the emotional journey is understood as an intrinsic dimension of the customer journey. The customer journey provides the temporal and interactional structure, while the emotional journey refers to the emotional responses and interpretations that emerge as users move through that structure. More specifically, the emotional journey refers here to the emotional shifts users experience before, during, and after exercise-related app use. These shifts may occur when a user opens the app, receives feedback, completes a workout, encounters a tracking problem, reviews progress, or returns after inactivity.

Emotions matter because they influence how users interpret experiences, form memories, evaluate trust, and describe their motivation or disengagement

(Palmer, 2010). In health and exercise applications, this is especially relevant. App use is often connected to bodily effort, self-discipline, recovery, progress, failure, and self-evaluation. A reminder, score, feedback message, or moment of silence may therefore carry more emotional weight than it would in a less personal service context.

From a corporate communication standpoint, this expanded understanding positions communication as part of value creation and relationship building. Communication is not only located in explicit messages. It is embedded throughout the user journey: in the tone of reminders, the framing of achievement messages, the clarity of feedback, the timing of prompts, and even in moments where the app does not respond. Villani (2018) argues that effective customer experience also depends on emotionally intelligent communication that resonates with users' needs, expectations, and values, alongside service quality and usability. In this view, app-based communication may influence satisfaction, trust, and loyalty-related perceptions (Jaakkola & Alexander, 2024; Villani, 2018).

This does not mean that every emotional response leads directly to continued use or disengagement. Rather, emotional responses are treated in this thesis as part of how users make sense of the app and its role in their exercise-related routines. For example, Bitrián et al. (2020) and Feng et al. (2020) show that gamification and progress tracking may initially motivate users, but their value depends on how they are experienced in context. A poorly timed or generic message may contribute to frustration, while well-framed feedback or guidance may support perceived motivation, trust, or willingness to continue app use. The emotional journey is therefore central to understanding how app-based communication becomes meaningful beyond its functional purpose.

Digital technologies add another layer to this process. Health and exercise applications can provide real-time interaction, personalised experiences, automated feedback, and adaptive communication. Märtin et al. (2023) suggest that emotion and behaviour tracking can support communication that responds dynamically to users' changing states. However, this thesis approaches such possibilities critically. Personalisation or automation is not assumed to be emotionally intelligent by default. Its significance depends on whether users experience communication as relevant, understandable, supportive, and respectful of their agency.

The customer journey also has a structural role in this study. Becker et al. (2020) introduce a goal-oriented perspective, highlighting how user motivations shape engagement patterns across the journey. Voorhees et al. (2017), in turn, propose a stage-based model that allows researchers to analyse emotional responses and satisfaction at different phases of interaction. In this thesis, the customer journey is not treated as a fixed commercial funnel. Instead, it is used as a flexible structure for locating emotionally significant app interactions before, during, and after exercise-related app use. This interpretation informed the eight-stage app-based customer journey scaffold used later in the interview design.

Together, customer journey and emotional journey perspectives provide the primary framework for the study. They make it possible to examine both where app-based interactions occur and how users emotionally interpret them. The following section turns to emotionally intelligent communication, which

provides the corporate communication lens for evaluating how app-based touchpoints may support, pressure, guide, validate, or distance users.

2.3 Emotionally intelligent communication and barriers to engagement

Emotional intelligence refers broadly to the capacity to perceive, understand, regulate, and use emotions in ways that support thinking, action, and social interaction (Goleman, 1995; Mayer & Salovey, 1997). In organisational contexts, emotional intelligence has also been conceptualised through interrelated competencies such as self-awareness, self-regulation, empathy, motivation, and social skills (Boyatzis et al., 1999; Goleman, 1998). These competencies have been widely discussed as resources for communication, leadership, decision-making, and relationship management (Hess & Bacigalupo, 2011; Sigmar et al., 2012).

In this thesis, emotionally intelligent communication refers to communication that is sensitive to the emotional, behavioural, and situational context of the user. It is not limited to encouraging or motivational language. Rather, it concerns how app-based messages, prompts, feedback, silence, timing, format, guidance, and interaction cues are shaped in relation to users' emotional readiness and interpretive context. In the field of corporate communication, this is relevant because digital touchpoints do not merely deliver information; they also shape how users perceive the organisation behind the service. In this sense, the application functions as an organisational communication interface through which users may infer reliability, care, competence, or neglect.

Research on digital corporate communication also shows that the perceived tone of mediated communication can shape how publics evaluate organisations. Dijkmans et al. (2015) found that conversational human voice mediated the relationship between exposure to a company's social media activities and perceived corporate reputation. Although their study examines social media rather than app-based communication, it supports the broader point that digitally mediated tone, responsiveness, and perceived human presence can influence how users interpret the organisation behind a service.

Drawing on Hess and Bacigalupo (2011), this thesis views emotional intelligence as a communicative competency that can help organisations respond to users in ethically and emotionally attuned ways. In health and exercise applications, such communication must balance organisational goals, such as engagement and continued use, with users' changing emotional states. A message that is intended to motivate may be experienced as supportive in one situation but pressuring or irrelevant in another. Emotionally intelligent communication therefore requires attention to timing, tone, context, and the user's sense of agency.

Automation and AI-mediated communication add another layer to emotionally intelligent app communication. Sundar's (2020) concept of machine agency helps explain why automated app communication can become relationally and emotionally significant: users may attribute competence, autonomy, or

intentionality to technological systems, especially when those systems recommend, evaluate, personalise, or guide action. From a corporate communication perspective, this is important because automated systems may increasingly act as interfaces between organisations and their stakeholders. Syvänen and Valentini (2020) show that conversational agents are already used in online organisation–stakeholder interactions, but that their organisational, stakeholder, and ethical implications remain underexamined in corporate communication research. Lindholm and Sihvonen (2024) similarly describe communicative AI as algorithmic systems with communicative agency, whose human-like language, interaction style, and perceived role can shape users' emotions, trust, and critical evaluation.

Agency and transparency are therefore especially important in AI-supported digital environments. Kang and Lou (2022) show that user engagement in AI-powered platforms is shaped through the relationship between human agency and machine agency. This helps explain why app-based communication should not be understood as passively received: users may accept automated feedback, but they may also question its logic, adapt it to their own goals, or limit the authority it has over their exercise-related decisions. Liu (2021) further shows that users' uncertainty about AI decisions can increase when decision-making appears to be governed by machine-generated rules, while meaningful explanations can reduce uncertainty and strengthen trust. Applied to health and exercise applications, these findings suggest that automated recommendations, scores, feedback, and AI-mediated guidance become more emotionally supportive when users can understand their basis and evaluate their relevance.

However, emotionally intelligent communication is not always easy to achieve in digital environments. Users may encounter barriers that inhibit engagement, including perceived risk, emotional discomfort, vulnerability, or resistance to behavioural change (Joachim et al., 2018). In health and exercise applications, these barriers may appear as avoidance, defensiveness, frustration, or disengagement when communication is poorly timed or emotionally misaligned. For example, a reminder after inactivity may encourage one user but trigger guilt or inadequacy in another.

Santos and Ponchio (2021) argue that emotional barriers, including discomfort, arousal, and diminished pleasure, can influence resistance to innovation in digital contexts. Although their study focuses on digital banking, the same broader dynamic is relevant to health and exercise applications, where communication often concerns bodily performance, discipline, progress, and self-evaluation. App-based prompts, scores, reminders, or feedback may therefore become emotionally charged when they are interpreted as judgemental, generic, intrusive, or poorly aligned with the user's situation.

Organisational and technological constraints may also limit emotionally intelligent communication. Evert et al. (2023) note that emotional intelligence can be challenged by digital mediation and structural conditions that reduce the richness of communication. In app environments, communication is often compressed into short prompts, automated messages, scores, or notifications. This can make it difficult to communicate nuance, empathy, or recognition. As a result,

even well-intended communication may appear cold, generic, or emotionally flat if it fails to acknowledge the user's context.

For communication professionals, these challenges underline the need to consider app-based communication at both strategic and micro-interactional levels. Emotionally intelligent communication should not rely on standardised motivational messages alone but should consider how users may interpret communication in specific moments of use. A user returning after inactivity may need reassurance rather than pressure; a user completing a difficult workout may value recognition rather than optimisation; and an experienced user may prefer transparent data over simplified encouragement.

In sum, emotionally intelligent communication is best understood as a responsive practice rather than a fixed communication style. It requires sensitivity to users' emotional states, behavioural context, and interpretive agency. This also points to the importance of emotional granularity: if communication is to respond intelligently to emotion, it matters how precisely emotions can be identified, named, and interpreted. The following section therefore turns to Plutchik's model as a practical tool for supporting the categorisation and discussion of reported emotional responses.

2.4 Emotion reflection, emotional granularity, and Plutchik's model

While the previous sections have discussed emotional journeys and emotionally intelligent communication, it is also important to consider how emotions can be recognised, named, and reflected upon in empirical research. This thesis relies on participants' own descriptions of emotionally significant app-related experiences. Therefore, the study does not treat emotions as directly measurable psychological states, but as reported, interpreted, and linguistically expressed experiences.

Emotional granularity refers to an individual's capacity to identify, differentiate, and describe affective experiences using specific emotional terms, such as proud, hopeful, discouraged, or relieved, rather than broad descriptors such as feeling good or bad (Tan et al., 2022). This concept is relevant for this study because participants differed in how precisely they were able to describe their emotional responses to app-based communication. Some emotions were easy to name, while others emerged through reflection, comparison, or discussion during the interview. These differences shaped how emotional experiences could be explored and interpreted.

This perspective aligns with Barrett's (2017) Theory of Constructed Emotion, which understands emotions as contextual, conceptual, and shaped through prior experience rather than as fixed universal reactions. From this view, emotions are not simply triggered by app features in a direct or mechanical way. Instead, users interpret bodily sensations, situational cues, previous experiences, and learned emotion concepts when making sense of what they feel. For this thesis, the key implication is methodological and communicative: users' reported

emotions are treated as interpreted accounts rather than objective measurements of internal states.

These insights are relevant for corporate communication because app-based communication may influence not only what users feel, but also how they make sense of and describe those feelings. Tone, wording, timing, feedback, and contextual cues can provide users with language and interpretive frames for understanding their experiences. For example, after a missed-workout notification, one user might describe feeling guilty, another disappointed, and another newly motivated. These distinctions matter because they suggest different communicative consequences: guilt may produce avoidance, disappointment may invite reflection, and motivation may support re-engagement.

In health and exercise applications, this issue is particularly important because app communication often concerns bodily effort, progress, recovery, discipline, and self-evaluation. Feedback, reminders, scores, and progress summaries may therefore become emotionally meaningful not only because of what they report, but because of how they frame the user's experience. This aligns with Yoon et al.'s (2016) argument that recognising nuanced emotional responses can support more emotionally resonant human-centred products and services. Emotionally intelligent communication can therefore support emotional clarity by helping users interpret their experiences without reducing them to overly simple categories or moralising judgements.

To support participants' emotion reflection and the structured categorisation of reported emotional responses, this study uses Plutchik's (1980) psychoevolutionary model of emotion, commonly known as the Wheel of Emotions. Plutchik originally grounded emotions in evolutionary theory, viewing them as adaptive mechanisms that may guide behaviour, such as fear being associated with withdrawal or anger with confrontation. In this thesis, however, the model is not used to explain how emotions are formed. Rather, it serves as a practical tool for supporting participants' reflection and organising reported emotional vocabulary during analysis. Its visual and categorical structure makes it useful as a supportive device for helping participants articulate emotional experiences during the interview process.

The model identifies eight primary emotions—joy, trust, fear, surprise, sadness, disgust, anger, and anticipation—which are arranged in oppositional pairs, such as joy versus sadness and anger versus fear. These core emotions can combine into more complex affective states, such as love, remorse, and optimism, and each exists in varying degrees of intensity. The figure used in this thesis is a simplified adaptation of Plutchik's structural model of emotions, particularly his multidimensional emotion model, circular arrangement of primary emotions, and discussion of emotion dyads (Plutchik, 1980, pp. 157–164). As illustrated in Figure 2 below, the circular layout helps visualise emotional intensity, contrast, and combination. In this thesis, the model is used to examine how reported emotions may intensify, weaken, or shift in response to different app-related events.

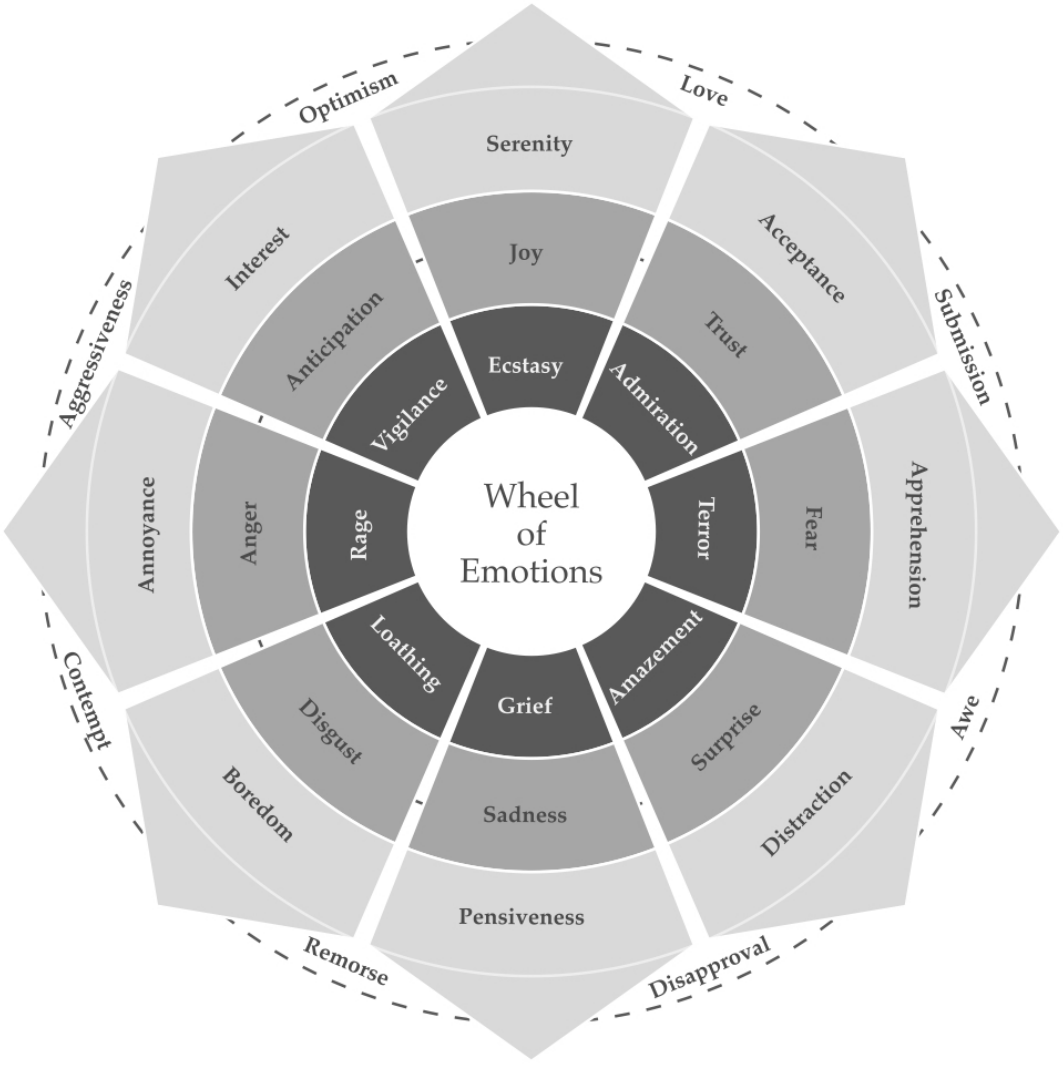


FIGURE 2. Plutchik's Wheel of Emotions, adapted from Plutchik (1980, pp. 157–164).

Plutchik's model supports this study in three main ways. First, it provides a predefined emotional vocabulary that can support consistency and clarity when participants describe their emotional responses during post-incident reflection. Second, the model offers a visual and intuitive tool that supports emotional awareness and expression among non-expert participants. Although it presents a simplified view of emotion, its accessibility makes it useful in an interview setting where participants may otherwise struggle to name or differentiate emotions. Third, the model helps organise reported emotions in terms of broad categories, intensity, and possible combinations, making it possible to compare emotional responses across app-related incidents without treating them as objective measurements.

In this thesis, the Wheel of Emotions is therefore used primarily as a methodological and analytical aid. It supports the discussion of emotional responses at key moments in the app interaction process, such as opening the app, choosing an activity, completing a workout, or receiving feedback. These moments

correspond to what Edvardsson and Roos (2001, p. 253) define as critical incidents: interaction points that users remember as unusually positive or negative. The Critical Incident Technique is discussed in more detail in Chapter 2.5.

From a corporate communication perspective, Plutchik's model offers a practical way to examine how different communicative actions may be associated with positive, negative, or mixed emotional responses. By analysing which communicative elements participants associated with emotions such as trust, relief, frustration, disappointment, or pride, the thesis can examine how app-based communication becomes emotionally consequential. For example, a well-timed and empathetic notification may reduce uncertainty and support trust, whereas poorly timed or impersonal communication may contribute to emotional dissatisfaction or disengagement (Bougie et al., 2003; Villani, 2018).

Used in this limited way, Plutchik's model supports emotion reflection and the comparison of participants' reported emotional responses without treating those responses as fixed, universal, or objectively measured emotional states. This distinction is important because the thesis also draws on constructionist perspectives to recognise emotions as contextual, linguistically mediated, and interpretive. The two perspectives are therefore used for different purposes: Plutchik supports practical emotion vocabulary and categorisation, while emotional granularity and constructed emotion help maintain a critical understanding of emotions as interpreted and communicatively mediated.

The next section builds on this discussion by examining the Critical Incident Technique, which was used to identify the emotionally salient app-use moments through which these reported emotional responses could be analysed.

2.5 Critical Incident Technique and emotionally significant touchpoints

Understanding which moments carry heightened emotional weight is essential for analysing how users interpret health and exercise applications. The Critical Incident Technique (CIT), introduced by Flanagan (1954), provides a structured approach for identifying incidents that are perceived as especially significant within a specific context. Although CIT was originally developed to study effective and ineffective work behaviours, it has since been widely adapted in service research to examine unusually positive or negative customer experiences (Edvardsson & Roos, 2001).

In this thesis, CIT is discussed as part of the literature review because it helps conceptualise emotionally significant service encounters. Its primary role, however, is methodological: it guided the identification and analysis of concrete app-use incidents that participants perceived as meaningful, emotionally charged, or consequential. This makes CIT particularly suitable for a study that examines not only what happened during app use, but how participants interpreted the emotional and communicative significance of those moments.

Contemporary qualitative applications of CIT have moved beyond Flanagan's original behavioural and functional orientation. Butterfield et al. (2005)

show that CIT has increasingly been applied to explore complex psychological constructs, subjective experiences, and cognitive-emotional linkages. Similarly, Watkins et al. (2022) emphasise the value of participant accounts in understanding the context, interpretation, and perceived consequences of critical incidents. This development supports the use of CIT in this study, where app-related incidents are analysed as situated and interpreted experiences rather than as objectively measurable events.

Edvardsson and Roos (2001, p. 253) define critical incidents as "interaction incidents, which the customer perceives or remembers as unusually positive or negative". This definition is useful for examining health and exercise applications, where emotionally significant moments may occur through small but meaningful interactions. A mistimed notification, unclear tracking failure, delayed feedback message, or supportive post-session summary may not appear important from a purely technical perspective. However, such moments can become critical when they affect the user's sense of control, competence, trust, motivation, or emotional connection with the application.

This is especially relevant in exercise-related contexts. Users may interact with applications while physically tired, focused, vulnerable, self-critical, or highly motivated. In such situations, minor communicative cues can carry emotional weight. A feedback message after a workout may validate effort, while a missing warning during tracking may create frustration or distrust. CIT therefore helps identify moments where app-based communication becomes emotionally and relationally meaningful.

From a communication perspective, critical incidents are not only shaped by explicit message content. They may also emerge through timing, silence, personalisation, ease of use, perceived empathy, or lack of acknowledgement. Villani (2018) similarly emphasises that customer experience is shaped by the wider interactional conditions through which users interpret a service. In this sense, CIT aligns with the broader communication perspective of this thesis: app-based touchpoints can become meaningful not only because of what the app says, but also because of how it responds, fails to respond, guides, interrupts, or enables action.

Skågeby's (2019) example of a frozen touchscreen payment device illustrates how a minor technological failure can become a critical incident when it disrupts control, interaction, and emotional confidence in a service situation. A similar logic applies to health and exercise applications. Small frictions such as failed tracking, unclear prompts, or delayed feedback may become emotionally significant when they interrupt the user's activity or weaken trust in the application. Conversely, small supportive cues, such as clear audio feedback or a meaningful post-session summary, may strengthen users' perceived motivation or confidence.

In this thesis, CIT therefore functions as a bridge between emotional journey theory and the analysis of app-based communication. It helps locate the specific moments in which communicative features become emotionally salient and allows the analysis to focus on concrete incidents rather than general opinions about applications. Together with stimulated recall and Plutchik's Wheel of Emotions, CIT supports the study's aim of examining how participants describe,

interpret, and emotionally evaluate app-based communication across the user journey.

The following section summarises the theoretical and methodological concepts that guide the study and clarifies how they are applied in the empirical analysis.

2.6 Summary of the theoretical framework

Figure 3 summarises the layered role of the main theoretical concepts and shows how they support the empirical analysis in this thesis. The customer journey and emotional journey perspectives draw primarily on Lemon and Verhoef (2016), Følstad and Kvale (2018), Palmer (2010), and Mele et al. (2021). Emotionally intelligent communication is informed by Goleman (1995, 1998), Mayer and Salovey (1997), Hess and Bacigalupo (2011), Sigmar et al. (2012), and Villani (2018), while the interactional view of communication draws on Watzlawick et al. (1967). The understanding of app-based communication as mediated organisational communication is further supported by corporate communication and public relations research on CommTech, digital organisation–stakeholder relationships, stakeholder engagement through mobile applications, and data-driven stakeholder insight (Brockhaus et al., 2022; Lock, 2019; Seiffert-Brockmann et al., 2018; Wiencierz & Röttger, 2017). In addition, research on conversational human voice, machine agency, user agency, transparency, conversational agents, and communicative AI supports the interpretation of automated and AI-mediated app communication as relationally and emotionally consequential (Dijkmans et al., 2015; Kang & Lou, 2022; Lindholm & Sihvonen, 2024; Liu, 2021; Sundar, 2020; Syvänen & Valentini, 2020). The constructionist and emotional granularity perspective is based on Barrett (2017), Tan et al. (2022), and Yoon et al. (2016). Plutchik's Wheel of Emotions and the Critical Incident Technique are positioned as methodological and analytical tools, not as the primary theoretical framework of the thesis (Butterfield et al., 2005; Edvardsson & Roos, 2001; Flanagan, 1954; Plutchik, 1980; Watkins et al., 2022).

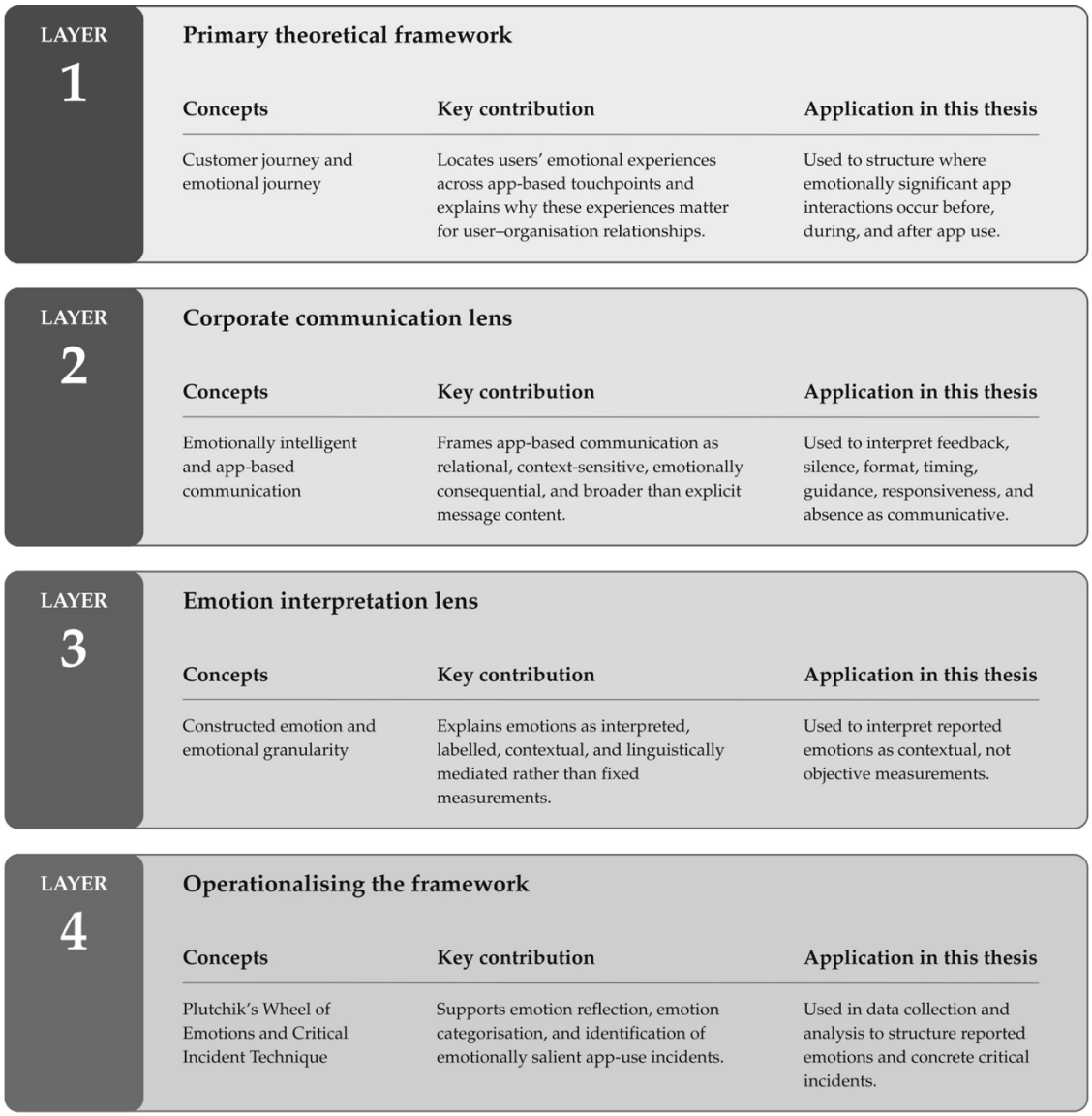


FIGURE 3. Layered theoretical framework of the study.

The figure should be read as a hierarchy rather than as a flat list of concepts. The customer journey and emotional journey provide the main structure for locating app-based experiences. The corporate communication lens supports the interpretation of feedback, silence, format, timing, guidance, responsiveness, and absence as communicative. This interpretation is also informed by the interactional view of communication, which broadens communication beyond explicit message content. Constructed emotion and emotional granularity support the interpretation of reported emotions as contextual and linguistically mediated rather than fixed measurements. Plutchik's model and the Critical Incident Technique support the empirical move from broad theoretical concepts to the identification, categorisation, and analysis of emotionally significant app-use incidents.

3 DATA AND METHODOLOGY

This chapter explains how the theoretical framework was translated into an empirical research design. The study investigates how users describe emotionally significant moments during their interactions with health and exercise applications, with particular attention to communicative touchpoints such as notifications, feedback messages, progress updates, reminders, and moments of silence or perceived absence of support. Operating within a contextualist epistemology, the research design is qualitative and interpretive; the aim is not to measure emotions as objective or quantifiable states, but to understand how users construct meaning and make sense of emotionally charged app interactions in their own words (Braun & Clarke, 2006).

Methodologically, this means that app features were not analysed as design objects in themselves. They were analysed only insofar as participants interpreted them as communicative cues. For example, silence, feedback, format, and personalisation logic were relevant not simply as technical or design features, but because participants described them as communicating support, uncertainty, authority, pressure, recognition, or absence. This interpretive focus anchors the analysis in corporate communication rather than in usability evaluation or psychological measurement.

The customer journey framework is used to situate these experiences across different moments of app use, while the Critical Incident Technique (CIT) supports the identification of interaction points that participants perceived as unusually meaningful, positive, negative, or emotionally salient. Stimulated recall, supported by participants' recent app use and app walkthroughs during the interviews, was used to anchor these accounts in concrete interaction moments rather than general attitudes.

Plutchik's (1980) Wheel of Emotions (Figure 2) provides a structured vocabulary for categorising reported emotional responses, whereas the concept of emotionally intelligent communication informs the interpretation of how app-based communication is perceived in relation to empathy, contextual sensitivity, and responsiveness to users' emotional needs. Together, these frameworks guide the collection, organisation, and interpretation of the empirical data.

Overall, this chapter outlines the methodological choices made to answer the research questions. It first presents the qualitative and interpretive research approach adopted in the study, followed by a description of the research data, participant recruitment, and interview procedure. It then explains how the Critical Incident Technique and stimulated recall informed the interview design, and how Plutchik's Wheel of Emotions was used to support the categorisation of reported emotional responses.

Finally, the chapter details the use of theoretical, latent, and reflexive thematic analysis as the analytical approach (Braun & Clarke, 2006; Maguire & Delahunt, 2017), and discusses ethical considerations and data protection, including the transparent use of artificial intelligence (AI) as a supportive aid (Xu, 2026). The broader evaluation of the study's trustworthiness, reflexivity, limitations, and implications is presented later in Chapter 5.

3.1 Research design and research questions

This thesis examines how communication can shape emotionally engaging user journeys in the context of health and exercise applications. Grounded in a multidisciplinary theoretical framework, the research does not concentrate on app usability or interface design per se, but rather on how communicative elements such as tone, timing, content, feedback, format, silence, and guidance contribute to users' emotional responses and perceived engagement. The goal is to explore communication as an emotional and strategic element of digital service interaction.

Operating within a contextualist epistemology, the study adopts a qualitative and interpretive research design. This approach is appropriate because the research aims to understand how users describe, interpret, and assign meaning to emotionally significant moments during app use, rather than attempting to measure emotions as fixed or objectively observable states. Qualitative interviews are appropriate for this study because they allow participants to describe app-related experiences from their own perspective and to explain how they interpret the meaning of emotionally significant interactions (Brinkmann & Kvale, 2018). Because contextualism acknowledges both material reality and subjective interpretation, the research design prioritises concrete, situated app-related moments over general attitudes towards health and exercise applications. This distinction is critical; broad evaluations—such as whether notifications are generally motivating or annoying—fail to capture the specific communicative cues, contexts, and interpretive assumptions through which emotional responses are formed. This interview-centred design was further supported by Finnish methodological literature on qualitative interviewing, particularly the practical and interactional perspectives outlined in *Tutkimushaastattelun käsikirja* (Hyvärinen et al., 2017).

The empirical design draws on the Critical Incident Technique (CIT) to identify moments that participants perceived as unusually positive, negative, meaningful, or emotionally charged (Butterfield et al., 2005; Watkins et al., 2022).

While traditionally a behavioural tool, contemporary constructivist applications of the CIT explicitly support the exploration of complex psychological constructs and cognitive-emotional linkages, making it a suitable method for examining affective responses (Butterfield et al., 2005). Stimulated recall (Törrönen, 2017) is used to support participants in reflecting on these concrete app-related experiences. In line with stimulated recall literature, interview prompts were used to support the recall of situated experiences (Lyle, 2003), while the customer journey framework provides a structure for locating these experiences across different stages of app interaction.

Plutchik's (1980) Wheel of Emotions provides a structured vocabulary for categorising reported emotional responses without treating them as objective psychometric measurements. Together, these methodological choices support the analysis of how communicative elements are interpreted, what emotional responses participants report in relation to them, and how these responses are perceived to connect with motivation, engagement, or disengagement.

The following sections outline the methodological tools and processes used to investigate these questions. This includes the use of CIT to identify emotionally charged touchpoints, stimulated recall to support detailed reflection on app interactions, and Plutchik's emotion categorisation model to structure reported emotional responses. The chapter later details the application of theoretical, latent, and reflexive thematic analysis to interpret users' accounts from a strategic corporate communication perspective.

3.2 Research data and data collection

The empirical data for this thesis consisted of qualitative interview material generated through stimulated recall interviews with users of health and exercise applications. In line with the scope of the thesis, these applications included apps used in relation to exercise, training, recovery, bodily self-monitoring, or exercise-related reflection. The interviews were designed to elicit detailed accounts of emotionally significant app-related moments, with particular attention to communicative touchpoints such as notifications, feedback messages, progress updates, guidance, app format, silence, and personalisation logic. Rather than collecting objective behavioural metrics, physiological emotion data, or real-time app analytics, the study focused on participants' own descriptions and interpretations of how app-based communication shaped their emotional experiences. This approach is consistent with an interview-based understanding of qualitative data, in which participants' accounts are treated as situated descriptions produced through interaction rather than as direct recordings of experience itself (Ruusuvuori & Tiittula, 2017).

Data collection was informed by the Critical Incident Technique (CIT), which guided participants to recall specific moments that they perceived as unusually positive, negative, meaningful, or emotionally charged. This incident-based approach was chosen because the research questions required more than general evaluations of health and exercise applications. Instead of asking

participants only whether they liked or disliked certain app features, the interviews encouraged them to describe concrete situations: what happened, what the app appeared to communicate, how they interpreted the interaction, what emotional response followed, and how the moment related to their motivation or engagement.

Stimulated recall was used to support this process by helping participants anchor their reflections in concrete app experiences rather than abstract opinions. The approach was informed by Törrönen's (2017) discussion of *virikehaastattelu*, or stimulus-based interviewing, in which stimuli are used to support reflection and elicit situated accounts of experience. In this thesis, participants' recent app experiences and, where possible, their own app interfaces functioned as the primary stimuli for reflection: participants were invited to revisit emotionally meaningful moments before, during, and after exercise-related app use. The customer journey framework provided a loose structure for this walkthrough, helping organise the conversation around different stages of interaction, such as opening the app, preparing for an activity, receiving guidance, completing a workout, reviewing feedback, reflecting on the experience, and considering future app use.

Plutchik's (1980) Wheel of Emotions was used as a supportive tool during data collection to help participants articulate and differentiate their reported emotional responses. The wheel was not used as a psychometric measurement instrument, but as a shared vocabulary for discussing emotions. Participants were able to select emotion terms from the wheel, describe their feelings in their own words, or combine both approaches. This was intended to support emotional granularity during recall while avoiding the assumption that participants' emotional experiences could be reduced entirely to predefined categories.

The interviews were conducted in Finnish to allow participants to express their experiences and emotions with nuance and comfort. The interviews were recorded and transcribed in Finnish using the MyJYU AI transcription tool, after which the transcripts were manually reviewed and corrected by the researcher. This review process was important because transcription is not a purely mechanical transfer of speech into text, but an interpretive process shaped by the aims of the study (Ruusuvuori & Nikander, 2017). In the findings chapter, selected illustrative quotations are translated into English by the author and checked against the original Finnish transcripts to preserve meaning, tone, and emotional nuance as accurately as possible. The following section describes the participant recruitment process, inclusion criteria, and sample characteristics.

3.3 Participants and sampling

Participants were recruited through convenience sampling from the researcher's personal, acquaintance, and university networks. This sampling strategy was considered appropriate for the exploratory and qualitative nature of the study, as the aim was not to produce statistically generalisable findings, but to gain in-depth insight into how users describe emotionally significant experiences with

health and exercise applications. The sample was therefore selected for its relevance to the research topic rather than for statistical representativeness.

The main inclusion criterion was that participants used at least one health- or exercise-related application and were able to discuss a recent or memorable experience with the app in detail. During recruitment, participants were informed that the study focused on emotional experiences and communication within health and exercise app use. All participants were Finnish speakers, which allowed the interviews to be conducted in Finnish and enabled participants to express emotional and experiential nuances in their first language.

The final sample consisted of six participants, pseudonymised as H1–H6. One interview, H1, was initially conducted as a pilot interview but was included in the final dataset because it followed the same overall interview protocol and produced data relevant to the research questions. The remaining five interviews were conducted as main interviews. Interview lengths ranged from 40 to 69 minutes. Participants were young adults aged between 24 and 30, and the sample included four female participants and two male participants. Table 1 summarises the basic participant characteristics.

TABLE 1. Participant characteristics.

Participant	**Interview type**	**Age group**	**Gender**	**Interview duration**
H1	Pilot	24–26	Female	44 min
H2	Main	24–26	Female	64 min
H3	Main	24–26	Female	69 min
H4	Main	24–26	Female	45 min
H5	Main	27–30	Male	49 min
H6	Main	27–30	Male	40 min

Participants varied in their app-use contexts and training backgrounds. One participant had used their selected app for less than three months, while the others had used their apps for more than twelve months. Most participants used their selected apps daily or several times per week. The applications discussed included yoga, running, fitness tracking, workout programming, and coaching-related applications. Their roles in participants' routines varied from tracking and feedback to routine creation and coach-mediated training support. Participants' training experience ranged from intermediate experience of approximately two to five years to expert-level experience of more than ten years. This variation was useful for the study because it allowed the analysis to consider both everyday app use and more experienced users' critical interpretations of app-based communication. Table 2 summarises the app-use context of each participant.

TABLE 2. App-use context.

Participant	App(s) discussed	App role	Usage frequency	Training experience
H1	Yin Yoga Bliss Lite	Routine creation	Weekly	Expert, 10+ years
H2	Apple Fitness	Tracker	Daily	Intermediate, 2–5 years
H3	RunKeeper	Tracker	Several times a week	Intermediate, 2–5 years
H4	Hevy; Suunto	Routine creation; Tracker	Less often; Several times a week	Advanced, 5–10 years
H5	Oura	Tracker	Daily	Expert, 10+ years
H6	TrainingPeaks	Coach-mediated support	Daily	Advanced, 5–10 years

Although the sample was small and not statistically representative, it was suitable for the aims of this qualitative study. In qualitative interview research, sample adequacy does not depend only on the number of participants, but also on the amount of information the sample provides in relation to the study aim, sample specificity, theoretical background, quality of dialogue, and analytical strategy. Malterud et al. (2016) describe this as information power: the more relevant information a sample holds for the study, the fewer participants may be needed.

In this thesis, the sample offered sufficient information power for several reasons. First, the study had a focused aim: to examine how users interpret emotionally significant app-based communication in health and exercise applications. Second, participants were selected because they had direct experience with relevant applications and could discuss concrete recent or memorable app-use situations. Third, the interviews were relatively detailed, lasting between 40 and 69 minutes, and were supported by stimulated recall, app walkthroughs, Plutchik's Wheel of Emotions, and Critical Incident Technique-informed questioning. Fourth, the analysis was not intended to produce broad population-level claims, but to develop an in-depth, incident-based interpretation of communicative and emotional mechanisms across six participant accounts and 36 critical incidents.

The sample therefore supported a detailed qualitative analysis of emotionally significant app-use experiences. Its adequacy should be understood in relation to the depth, relevance, and specificity of the data rather than through statistical representativeness or saturation in a quantitative sense.

3.4 Interview design and procedure

The interviews were designed as semi-structured stimulated recall interviews, drawing on qualitative interview methodology that emphasises participants' own meaning-making while allowing the researcher to guide the conversation

through theoretically relevant themes (Brinkmann & Kvale, 2018; Hyvärinen et al., 2017). This format allowed the researcher to follow a consistent structure across participants while retaining flexibility to ask follow-up questions and pursue emotionally significant moments in greater depth. The interview design combined three methodological elements: a customer journey-based walkthrough, the use of Plutchik's Wheel of Emotions as an emotion-reflection aid, and Critical Incident Technique-based deepening questions. Together, these elements supported the collection of detailed accounts of how participants interpreted app-based communication before, during, and after health- or exercise-related app use. The semi-structured interview guide, including the app walkthrough and Critical Incident Technique questions, is included as Appendix 2.

Before each interview, participants received information about the study, data handling, and consent. At the beginning of the interview, consent was confirmed again, and participants were reminded that the interview would be recorded. They were also informed that they could decline to answer any question, pause the interview, or withdraw from participation. Participants were asked to have access to the health or exercise application discussed in the interview, whether on a phone, smartwatch, or another relevant device. In line with stimulus-based interview logic, the application functioned as a stimulus for recall, helping participants return to concrete moments of app use, thereby reducing reliance on abstract memory and broad evaluations of the application (Lyle, 2003; Törrönen, 2017).

The interviews began with background questions concerning the participant's app use. These questions explored which application the participant used, the duration and frequency of their usage, the role the app played in their routine, and the specific exercise experience discussed during the interview. The opening section helped establish the context of app use and clarified whether the application functioned primarily as a tracker, routine creation tool, coaching platform, or source of feedback and guidance.

After the background section, participants were asked to describe a recent health or exercise app experience related to exercise, training, recovery, self-monitoring, or reflection. The interview then followed a loose customer journey structure based on an eight-stage app-based customer journey scaffold developed for this study. The stages were: pre-use trigger and entry; session set-up and intention-setting; in-session interaction and micro-communications; immediate post-session feedback and appraisal; reflection and meaning-making; social interaction and external validation; next-step prompts and retention nudges; and exit, aftercare, and re-entry patterns. This structure was not used as a rigid sequence, as the applications and user routines differed across participants. Rather, it served as a flexible scaffold for locating emotionally significant moments across the app-based journey.

Plutchik's (1980) Wheel of Emotions (Figure 2) was introduced during the interview as a supportive tool for emotional reflection. Participants were invited to use the wheel when describing their emotional responses, but they were not required to select only from the predefined emotions. They could also describe their feelings in their own words or combine their own descriptions with terms from the wheel. When relevant, participants were also asked to describe the

perceived intensity of an emotion, for example as mild, moderate, or strong. The purpose of this tool was to support emotional granularity and reflection, not to measure emotions psychometrically.

The latter part of the interview focused on emotionally significant incidents. Drawing on the Critical Incident Technique, which is used to identify significant incidents and their perceived consequences, participants were asked to identify moments that felt especially positive, negative, frustrating, motivating, meaningful, or otherwise emotionally salient (Butterfield et al., 2005; Flanagan, 1954; Watkins et al., 2022). These incidents were then explored through follow-up questions concerning what happened, what the app appeared to communicate, how the participant interpreted the moment, what emotional response followed, and whether the incident affected their motivation, engagement, or intention to continue using the app. This structure helped connect communicative elements, emotional responses, and perceived behavioural consequences.

The interviews also included reflection-oriented questions concerning how the app could have communicated differently in emotionally significant moments. These questions were included to support the strategic communication aim of the thesis: to understand not only how users experienced app-based communication, but also how such insights could inform more emotionally intelligent communication strategies. Participants were therefore encouraged to reflect on what kind of tone, timing, guidance, feedback, or support would have felt more appropriate in specific situations.

The interviews were conducted either in person or remotely via Zoom, depending on practical arrangements and participant availability. H1, H4, and H5 were interviewed in person, while H2, H3, and H6 were interviewed via Zoom. Both formats followed the same interview structure and relied on the same interview guide. In remote interviews, participants referred to their application verbally and, when relevant, described the interface or app experience in detail. Some participants also showed aspects of the applications to the camera. In in-person interviews, participants could also show parts of the application to the researcher when they felt comfortable doing so. Across both formats, the central aim remained the same: to support participants in recalling and interpreting concrete app-related moments as situated emotional and communicative experiences.

3.5 Data analysis

The data were analysed using a theoretical, latent, and reflexive approach to thematic analysis. Thematic analysis was considered suitable for this study because it allows the researcher to identify patterned meanings across qualitative data while remaining flexible enough to accommodate both theoretically guided and data-driven interpretation (Braun & Clarke, 2006; Maguire & Delahunt, 2017). In this thesis, the analysis was theoretical because it was guided by the research questions and the conceptual framework introduced in Chapter 2, particularly the customer journey, emotionally intelligent communication, Critical Incident

Technique, and Plutchik's Wheel of Emotions. It was latent because the analysis did not focus only on what participants explicitly said; rather, it sought to interpret the underlying assumptions, meanings, and communicative logics through which participants made sense of app-based interactions. It was reflexive because theme development required active interpretation by the researcher rather than mechanical code-counting.

The analysis began with familiarisation, during which the Finnish-language interview transcripts were read repeatedly. After transcription and manual checking, each transcript was read several times to gain an overall understanding of the participant's app-use context, emotional experiences, and communicative interpretations. During this stage, initial notes were made concerning emotionally significant moments, recurring app-related touchpoints, and potential links to the research questions. Attention was paid to moments where participants described a shift in emotion, motivation, trust, frustration, engagement, or disengagement.

The next stage involved identifying and extracting relevant incidents from the transcripts. In line with the Critical Incident Technique, the analysis focused on moments that participants described as unusually positive, negative, meaningful, frustrating, motivating, or emotionally salient. These incidents were not limited to explicit messages or notifications. They also included broader communicative cues, such as app format, silence, missing guidance, personalisation logic, feedback structures, visual presentation, or perceived absence of support. This approach reflected the broader communicative perspective of the thesis, in which both what the app communicates and what it fails to communicate may shape emotional experience.

The extracted material was organised into a structured analysis workbook. Each entry included the participant ID, app context, micro/macro classification, relevant customer journey phase, touchpoint or anchor, short description of the incident, contextual factors, data extract, communicative element, reported emotion, Plutchik-based emotion category, valence, intensity where available, behavioural implication, improvement implication, and analyst memo.

In total, the final incident-based dataset contained 36 critical incidents extracted from the six interviews. Of these, 19 were coded as micro-level incidents, referring to situated moment-by-moment app touchpoints, and 17 were coded as macro-level incidents, referring to broader or longer-term communicative observations about the user–app relationship. These numbers were used to support transparency and traceability in the analysis, not to produce statistical generalisations.

The distinction between micro-level and macro-level incidents helped structure the analysis. Micro-level incidents referred to concrete, situated moments within app use, such as receiving feedback, encountering friction, or interpreting a specific app feature. Macro-level incidents referred to broader, recurring interpretations of the app's communicative logic, such as patterns of silence, trust, autonomy, or transparent app logic. This distinction supported comparison between immediate emotional reactions and more accumulated app-related interpretations.

Coding proceeded through an abductive logic, moving between theory-informed categories and data-driven observations. The initial coding was guided by the research questions and theoretical framework, including customer journey stages, communicative elements, CIT-informed incidents, and Plutchik-based emotion categories, while remaining open to patterns that were not predetermined by the theory. The main communicative coding categories included elements such as content, feedback, format, timing, guidance, silence, and personalisation logic, some of which were refined during the analysis. Plutchik's (1980) Wheel of Emotions was used to support emotion categorisation by linking participants' own emotion words to broader emotion categories, such as joy, trust, anger, surprise or anticipation. However, participants' own expressions were retained alongside Plutchik-based categories to avoid reducing their emotional accounts to predefined labels. Where relevant, emotional valence and intensity were also noted, as well as the participant's perceived behavioural response, such as continued use, avoidance, increased motivation, compensation, or disengagement.

After incident-level coding, the analysis moved towards theme development. Codes and incidents were compared within and across participants to identify recurring patterns in how app-based communication shaped emotional experience. This involved examining similarities, contrasts, and tensions between participants' accounts. For example, silence could be interpreted negatively when it left the user unsupported, but positively when it protected autonomy or reduced interruption. Similarly, post-session data could function not merely as information, but as emotional validation of physical effort. These interpretive comparisons helped move the analysis from descriptive coding towards more conceptual themes.

Candidate themes were then reviewed in relation to the full dataset, the coded incidents, and the research questions. Themes were refined by assessing whether they captured meaningful patterns across participants while remaining grounded in specific incidents and quotations. The final themes were selected not only because they appeared repeatedly, but because they explained important communicative and emotional mechanisms within app-based user journeys. This was particularly important because the study aimed to understand how communication shapes emotionally engaging user journeys, rather than simply listing app features that participants liked or disliked.

The analysis resulted in six final themes, which are presented in Chapter 4. Together, these themes address how users interpreted app-based communication, how emotional responses emerged across different stages of interaction, and how these responses were perceived to connect with engagement, motivation, or disengagement.

The final movement from codes to themes was iterative rather than linear: incidents were coded, compared, grouped, reconsidered, and refined in relation to both the research questions and the original transcript context.

The analysis process is summarised in Table 3. The table outlines the main analytical stages, the actions taken at each stage, and the purpose of each step. It is intended to clarify how the analysis moved from familiarisation with the

Finnish-language transcripts to incident identification, coding, theme development, and final theme refinement.

TABLE 3. Overview of the analysis process.

Stage	Analytical action	Purpose
1. Familiarisation	Finnish transcripts were read repeatedly after transcription and manual checking.	To understand each participant's app-use context, emotional experiences, and communicative interpretations.
2. Incident identification	Emotionally significant app-related moments were identified using CIT-informed logic.	To locate concrete moments where communication appeared to shape emotional experience.
3. Workbook organisation	Incidents were organised by participant, app context, journey phase, communicative element, emotion, behavioural implication, and analyst memo.	To create a structured and traceable basis for comparison.
4. Coding	Incidents were coded abductively using both theory-informed and data-driven categories.	To connect empirical material with the research questions and theoretical framework while remaining open to unexpected patterns.
5. Theme development	Codes and incidents were compared within and across participants.	To identify broader communicative and emotional mechanisms.
6. Theme refinement	Candidate themes were reviewed against the full dataset, research questions, and selected quotations.	To ensure that final themes were analytically meaningful and grounded in the data.

In addition to the stages summarised in Table 3, AI tools were used as supportive analytical aids during parts of the analysis process, in line with emerging discussions on the reflexive and ethical use of generative AI in thematic analysis (Xu, 2026). In practice, AI tools were used to support the organisation of coding ideas, comparison of coded incidents, reflection on possible alternative interpretations, and testing the clarity of emerging theme labels. They were also used to support reflection on whether candidate themes appeared coherent across subsets of the data, such as micro-level incidents, macro-level incidents, and incidents grouped by main communicative element. AI tools were not used to generate data, replace transcript reading, code the dataset autonomously, or make final analytical decisions.

AI-generated suggestions were treated as provisional prompts for reflection rather than as findings. All suggestions were checked against the original Finnish transcripts, the structured analysis workbook, researcher memos, and the research questions. Suggestions that were unsupported, overly general,

inconsistent with the data, or insufficiently aligned with the study's corporate communication focus were rejected or revised. The final coding decisions, theme definitions, theme names, interpretations, quotation selection, and final analytical claims were made by the researcher.

The ethical implications of this AI-supported process are addressed in the following section and in Appendix 1, while its trustworthiness and reflexivity implications are discussed in Chapter 5.

3.6 Research ethics and data protection

This study was conducted in accordance with the ethical principles for research with human participants outlined by the Finnish National Board on Research Integrity TENK (Tutkimuseettinen neuvottelukunta, 2023), as well as the University of Jyväskylä's guidance on student research data management and data protection (University of Jyväskylä, n.d.). TENK identifies informed consent as a central ethical principle in research with human participants and emphasises participants' rights, privacy, and responsible processing of personal data. JYU's student guidance similarly notes that thesis research involving living persons generally involves the processing of personal data and therefore requires attention to data protection, participant information, and responsible data handling.

Ethical and data protection considerations were addressed during the planning stage of the thesis. As part of the researcher's studies, a data management plan was prepared in connection with the course *Aineistonhallinnan perusteet* (Basics of Research Data Management). This plan, together with JYU guidance on data management and storage, informed the practical handling of the interview recordings, transcripts, and analysis files. JYU guidance states that a data management plan should describe a credible storage and processing environment, and JYU lists Microsoft Office365, including OneDrive, among its primary collaboration and storage tools for university accounts.

A preliminary Data Protection Impact Assessment screening was also completed. Based on this screening, the low-risk nature of the study, and discussions with supervisors, formal ethical review by an ethics committee was not required for this study. The study did not involve minors, vulnerable participants, medical intervention, deception, large-scale processing, systematic monitoring, automated decision-making, or profiling. The only DPIA screening item marked as applicable concerned the use of new technology, due to the use of AI-supported tools in the research process. The study did not involve the intentional collection of special categories of personal data, such as clinical health data, medical diagnoses, physiological measurements, or health records. Because data concerning health can fall under special categories of personal data, particular care was taken to keep the interview focus on participants' app use and emotional experiences rather than on medical or clinical information.

Participation in the study was voluntary. Before each interview, participants received information about the purpose of the study, the nature of participation, recording, data handling, and their rights as participants. At the

beginning of each interview, consent was confirmed again. Participants were informed that they could decline to answer any question, pause the interview, or withdraw from participation. The interviews focused on app-related experiences and emotional interpretations, and participants were not required to disclose private health information beyond what they considered relevant to their own app use.

The study involved the processing of personal data because interview recordings and transcripts may contain directly or indirectly identifiable information. To protect participants' privacy, all participants were pseudonymised using participant codes H1–H6. Identifying details, such as names, unnecessary location references, or other contextual details that could increase identifiability, were removed or generalised during transcript preparation and reporting. In the thesis, quotations are reported using participant codes only.

The interviews were recorded and transcribed in Finnish using the MyJYU AI transcription tool. After transcription, the researcher manually reviewed and corrected the transcripts against the recordings. This step was necessary both for analytical accuracy and for privacy protection, as automated transcription may misrecognise words, names, app-related terminology, or emotionally nuanced expressions. During this review process, transcripts were also pseudonymised and identifying details were removed before analysis. The use of MyJYU AI transcription was limited to producing initial transcripts; it did not replace the researcher's responsibility for checking, correcting, pseudonymising, and interpreting the data.

Research data were stored securely and accessed only by the researcher. The primary storage location was the researcher's university OneDrive, and the stored data were encrypted. During analysis, some working files were also stored on the researcher's personal computer, which was password-protected, encrypted, and accessible only to the researcher. Working files, including the analysis workbook, used participant codes rather than participants' real names. The data were not shared with external parties in identifiable form.

AI tools were used transparently and only as supportive aids during the research process. In transcription, the MyJYU AI transcription tool supported the production of initial Finnish transcripts, which were manually checked by the researcher. In analysis, generative AI tools, including ChatGPT, Gemini, and NotebookLM, were used to support organisation, comparison, summarisation, and reflection on coding and theme development. Where external AI tools were used, the material was pseudonymised and stripped of directly identifying details before processing. These tools were not used to generate raw data, make final analytical decisions, or replace the researcher's interpretation of the transcripts. AI-generated suggestions were treated as provisional prompts and checked against the original data, the analysis workbook, and the research questions. This approach follows emerging discussions on the need for transparency, reflexive awareness, and human accountability when using generative AI in qualitative analysis (Xu, 2026).

The reporting of findings was guided by the principle of protecting participants' confidentiality while preserving the meaning of their accounts. Selected quotations were translated from Finnish into English by the author. Translations

were checked against the original Finnish transcripts to maintain the participant's intended meaning, tone, and emotional nuance as accurately as possible. Where necessary, minor details were removed or generalised to reduce identifiability without altering the analytical meaning of the quotation.

Overall, the ethical aim of the study was to respect participants' autonomy, protect their privacy, and handle the interview material responsibly throughout the research process. These measures were especially important because the study focused on emotionally meaningful experiences, some of which involved frustration, vulnerability, motivation, self-evaluation, or perceived failure in relation to exercise and wellbeing.

4 RESEARCH FINDINGS

This chapter presents the findings of the thematic analysis. The purpose of the chapter is to show how participants interpreted emotionally significant moments in health and exercise application use, and how these interpretations were connected to app-based communication, emotional responses, motivation, and engagement. The findings are based on six stimulated recall interviews and the incident-based analysis described in the methodology chapter.

The results are organised around six themes developed through theoretical, latent, and reflexive thematic analysis. These themes do not represent isolated categories but interconnected communicative mechanisms through which participants made sense of app-based experiences. Across the data, communication was not limited to explicit messages such as notifications, feedback, or guidance. Instead, participants also interpreted format, silence, data presentation, app logic, and mediated coaching as meaningful communicative elements.

The chapter begins with an overview of the six final themes. The following sections then present each theme in detail, supported by theme-specific tables, translated interview excerpts, and examples from the analysed incidents. The chapter concludes with a summary of the main critical incident types identified across the final themes, clarifying how the CIT-informed incident analysis supported the thematic findings presented in this chapter.

4.1 Overview of the findings

This section provides an overview of the six themes developed through the thematic analysis. The themes describe how participants interpreted app-based communication and how these interpretations shaped their emotional experiences, motivation, and engagement. In this sense, app communication operated both through what the application explicitly communicated and through how the application structured, timed, withheld, or personalised interaction.

The findings should be read through the communicative lens established in the theory and methodology chapters. The themes developed from the analysis show that communicative elements such as format, silence, data presentation, and app logic were not experienced merely as technical features, but as meaningful signals within the app-mediated relationship between the user and the application. For example, a difficult interface could communicate indifference; silence could communicate either respect or abandonment; opaque personalisation could communicate misunderstanding; and feedback data could communicate recognition of bodily effort. In this sense, the findings concern not app design as such, but the emotional consequences of app-based communication.

As described in the methodology chapter, the interviews were supported by an eight-stage app-based customer journey scaffold. This scaffold helped locate emotionally significant incidents across different moments of app use, including set-up, in-session interaction, post-session feedback, reflection, and next-step planning. However, the final themes were not distributed evenly across all journey stages. Instead, the most analytically visible incidents clustered around moments where users had to interpret app communication, respond to feedback, rely on automated logic, or evaluate the app's role in their exercise routine.

Table 4 provides a concise overview of the six final themes. The purpose of the table is to orient the reader to the main communicative focus and emotional or engagement-related implication of each theme. The detailed development from reduced expressions to subthemes and main themes is presented in the theme-specific tables in Sections 4.2–4.7. A separate summary of the critical incident types identified in the analysis is presented at the end of the findings chapter.

TABLE 4. Overview of the final themes.

Theme	Communicative focus	Main emotional/engagement implication
The meta-communicative role of format: system friction versus structural support	How the structure, responsiveness, and accessibility of the app shaped the user's experience of support or friction.	Smooth format supported flow and reduced emotional effort, while rigid or unresponsive format created frustration and disrupted engagement.
The contextual duality of silence: respect and abandonment	How the absence of communication was interpreted differently depending on the user's situation and expectations.	Silence could protect autonomy and focus, but it could also create uncertainty, frustration, or a sense of being unsupported.
Cognitive offloading and the need for transparent app logic	How automated planning, recommendations, and app logic reduced or increased users' interpretive effort.	Automation created relief when it was understandable and relevant, but opaque logic produced doubt and reduced trust.
Post-session data as bodily validation	How feedback, summaries, scores, and progress data made physical effort visible and interpretable.	Post-session data validated effort, strengthened motivation, and sometimes repaired earlier negative emotions.
Humanised coaching and mediated authority	How app communication resembled, simulated, or mediated coaching-like support.	Coaching-like communication increased motivation when it felt timely, competent, relational, and trustworthy.
Expert user agency: auditing and emotional distancing	How experienced users evaluated, questioned, limited, or adapted app-generated communication.	Users preserved agency by auditing app logic, relying on bodily knowledge, and emotionally distancing themselves from metrics when needed.

4.2 Theme 1: the meta-communicative role of format: system friction versus structural support

The first theme concerns the meta-communicative role of app format. In this study, format refers to the structural conditions through which app communication becomes accessible, usable, or obstructive. This includes interface responsiveness, interaction flow, editability, access to controls, visual structure, and the extent to which the app's format supports the user's physical and emotional context. The findings suggest that participants did not experience format as a neutral

technical layer. Instead, format was interpreted as signalling something about the app's attentiveness, competence, and user-centredness.

Table 5 summarises how the theme was developed from reduced expressions into two subthemes and the final main theme. The table is not intended to quantify the findings, but to clarify how concrete app-use incidents were interpreted as part of a broader communicative pattern.

TABLE 5. Development of theme 1: the meta-communicative role of format.

Reduced expressions / codes	**Subthemes**	**Main theme**
Unresponsive hardware-software interface during physical activity Forced manual interaction interrupting physical flow Premature or rigid automated timing Inability to edit flawed historical data Missing scalable movement alternatives	System friction and structural rigidity	The meta-communicative role of format: system friction versus structural support
Seamless access to core actions Frictionless background automation Contextually appropriate audio or interface delivery Interface flow matching the user's physical state	Format as structural and situational support	

This theme was particularly visible in in-session interaction and post-session feedback. During exercise, participants were often physically engaged, time-sensitive, or focused on bodily performance. In such moments, even small structural frictions could become emotionally significant because they interrupted the user's flow or forced attention away from the activity. Conversely, when the app's format worked smoothly, it often became almost invisible. This invisibility was not meaningless; rather, it functioned as structural support by allowing the user to continue the activity without unnecessary cognitive or emotional effort.

A clear example of system friction appeared in H2's account of using a smartwatch during swimming. The participant described a situation where the screen failed to respond when they attempted to pause or control the workout during exercise. The problem was not simply that the interface malfunctioned, but that it malfunctioned in a context where the participant's hands and screen were wet, making responsiveness especially important. As H2 described:

> H2: "The screen didn't react to it... it was frustrating. I had just received that good feedback... and then I thought I could have logged a completely perfect workout there."

This incident illustrates how format can become communicative when it fails to accommodate the user's embodied context. In a swimming session, the user cannot interact with the app in the same way as when sitting still with dry hands. The unresponsive screen therefore communicated a lack of situational fit: the app or device appeared insufficiently adapted to the conditions in which it was being used. Emotionally, this created frustration and annoyance because the participant's attention was pulled away from the exercise itself and redirected towards solving a technical problem. In this sense, system friction disrupted both bodily flow and emotional engagement.

A similar form of friction appeared in H3's account of post-session data editing. The participant described forgetting to stop tracking a run at the correct time, after which the app's format did not allow the session to be corrected in the desired way. This made the recorded data less useful for comparison and weakened the motivational value of the feedback. H3 explained:

> H3: "It leaves you feeling pretty annoyed that you can no longer compare this run... it would be exactly what you need at that point [if a mistake happens]."

Here, the emotional response was not caused by the workout itself, but by the app's structural rules after the workout. The inability to edit or correct the data meant that a simple human mistake remained embedded in the user's progress history. This matters because post-session data was not experienced merely as information; it contributed to how the user interpreted effort, progress, and achievement. When the format prevented correction, it weakened the credibility and usefulness of the app's feedback. The app's structure was therefore interpreted as rigid rather than supportive.

H1's experience with a yoga application highlighted a different form of structural friction: the disruption of physical flow through forced manual interaction. During a yoga routine, the participant expected the application to guide them seamlessly. However, the app's format lacked automatic transitions or audio cues, forcing the user to manually interact with the device between poses:

> H1: "[...] no one there gives you the next movement, instead you have to open the phone yourself, open the app and look at what the next movement was. So, in a way, no one speaks to you there, saying hey..."

This example shows that format can become communicatively significant through what it implicitly expects of the user. Because the application required

active, manual engagement rather than functioning as a supportive background guide, it broke the user's physical immersion. For the participant, this created both practical and emotional disruption: they either had to interrupt their physical flow to handle the device or continue without clear guidance. From a communication perspective, the absence of an adaptive, hands-free structure can be understood as a communicative absence. It suggested that the app format was not sufficiently aligned with the continuous, embodied nature of a yoga session, transforming the format from a source of support into a source of friction.

However, the theme was not only negative. H3 also described a positive case in which the app's format supported the user precisely because it required little effort. When the participant needed to pause or stop a run quickly, a lock-screen widget made this easy without requiring them to unlock the phone or navigate through the app. The participant contrasted their own disappointment about physical stamina with appreciation for the app's functionality:

> H3: "A slight disappointment [in my own stamina]... but from the app's perspective, I'd say it's just a really good feature, the fact that it's fast and easy."

This example is important because it shows how structural support can buffer negative emotion. The participant's disappointment was directed towards their own physical performance, not the app. The app's format, by being fast and easy, did not add further emotional burden to an already vulnerable moment. Instead, it protected the user's autonomy and allowed them to manage the situation smoothly. In this sense, good format communicated care indirectly: not through encouraging words, but by reducing friction at the exact moment when the user needed ease.

Taken together, these incidents show that format operated as a form of meta-communication. It shaped what users could do, how easily they could act, and whether the app felt aligned with their physical and emotional context. When format was restrictive, unresponsive, or insufficiently adaptive, it signalled friction and sometimes a lack of user-centredness. When format was smooth, accessible, and responsive to the user's situation, it communicated structural support. This theme therefore extends the understanding of app-based communication beyond explicit messages by showing how the architecture of interaction itself can become emotionally meaningful. In health and exercise applications, communication is therefore also embedded in the way interaction is structured, enabled, or obstructed.

4.3 Theme 2: the contextual duality of silence: respect and abandonment

The second theme concerns the contextual duality of silence in app-based communication. In this study, silence refers not only to the complete absence of messages, but also to missing guidance, lack of warning, absence of feedback, non-interruption, or moments where communication was withheld or absent. Table 6

summarises how reduced expressions related to silence were developed into two subthemes: silence as abandonment, neglect, or lack of support, and silence as respect for autonomy and focus.

TABLE 6. Development of theme 2: the contextual duality of silence.

Reduced expressions / Codes	**Subthemes**	**Main theme**
Missing transition or safety guidance Uncommunicated tracking failures Absence of expected encouragement during physical strain Passive recording without interpretive response	Silence as abandonment or lack of support	The contextual duality of silence: respect and abandonment
Absence of commercial interruptions Non-intrusive app environment Withholding performance-heavy feedback after reflective activity Freedom from unnecessary prompts during focused activity	Silence as respect for autonomy and focus	

The findings suggest that silence was not experienced as neutral. Rather, participants interpreted silence through the context in which it occurred. When support, guidance, or recognition was needed, silence could be experienced as abandonment or lack of support. However, when users wanted autonomy, calm, or uninterrupted activity, silence could be experienced as respect.

This theme was particularly visible in in-session interaction, reflection, and app-use situations where participants expected either guidance or restraint. During exercise, participants often expected the app to provide timely support, especially when something went wrong or when guidance was required to continue the activity. In these moments, silence could create frustration, uncertainty, or a sense that the app had failed to recognise the user's situation. However, in other moments, especially during reflective or self-directed activity, the absence of communication could be experienced positively because it allowed the user to remain in control without unnecessary interruption.

A clear example of silence as abandonment appeared in H1's experience with a yoga application. The participant described a situation where the app did not provide a sufficiently coherent or supportive sequence between movements. Instead of actively guiding the user through a manageable progression, the app offered no communicative support during critical transitions, making the routine feel too demanding without enough intermediate support:

> H1: "I maybe didn't really find a continuum here… I just did my own intermediate movements in between… It felt like it wasn't even safe to do these, because they kind of went off the deep end."

This example shows how silence can become emotionally significant when it appears as missing guidance. The problem was not only that the app did not speak or provide more instruction, but that the absence of supportive transition cues left the participant responsible for filling the gaps themselves. In a yoga context, where bodily positioning, progression, and safety matter, silence was interpreted as insufficient support for the user's embodied situation. The participant's need to create their own intermediate movements suggests that the absence of guidance shifted responsibility from the system to the user. Rather than supporting flow, the lack of guidance created uncertainty and made the participant question whether the movements were safe.

A second example of negative silence appeared in H3's experience with RunKeeper. The participant described a situation where the app failed to track the run properly, but this problem was not clearly communicated during the activity. The issue only became visible afterwards, when the recorded distance did not correspond to the actual run:

> H3: "Then it just gave me that time… but then it showed that I'd only run 0.51 kilometres… it left me with an annoyed feeling."

Here, silence functioned as a lack of warning. The participant did not need constant motivational messaging, but they did need to know when the core tracking function had failed. Because the app remained silent at a moment when corrective action might still have been possible, the participant was left with inaccurate data and an annoyed emotional response. This incident also shows how silence can affect trust. In exercise applications, tracking data often functions as evidence of effort and progress. When the app fails silently, the user may not only lose useful data but also begin to question the reliability of the app's communication system.

However, silence was not always experienced negatively. H1 also described a contrasting situation in which the absence of direct feedback after Yin yoga was considered appropriate. In this case, the participant did not expect the app to provide an external evaluation, because the meaning of the activity was understood as more internally reflective:

> H1: "The app itself doesn't give me the answer… on the other hand, I don't really expect that from Yin yoga… it's maybe more of an internal feedback."

This example shows that silence can function as respect when it aligns with the emotional and experiential purpose of the activity. Unlike the earlier yoga example, where missing guidance created uncertainty during movement, this post-session silence was not experienced as abandonment. Instead, it supported the reflective nature of Yin yoga, where the participant did not expect the app to define the value of the experience. The absence of external evaluation allowed the user's own bodily and emotional interpretation to remain central. In this context, silence communicated restraint rather than neglect.

H3's account of an ad-free and non-intrusive app experience similarly showed silence as respect for autonomy. The participant appreciated that the app did not interrupt the workout with irrelevant messages or advertisements:

> H3: "You don't get constant annoying ads popping up here that you have to turn off in the middle of a workout… Those kinds of apps go straight in the trash."

This quote demonstrates that emotionally intelligent communication does not always mean adding more messages, prompts, or feedback. In this case, non-communication was valued because it protected the user's focus during exercise. The participant's strong rejection of apps that interrupt workouts shows that intrusive communication can quickly damage the user relationship. By contrast, the absence of advertisements or irrelevant prompts communicated respect for the user's attention and activity. Silence, here, was not emptiness; it was a form of boundary-respecting communication.

Taken together, these incidents show that silence operated as a contextual form of communication. When participants needed guidance, warnings, feedback, or encouragement, silence was experienced as abandonment, lack of support, or missed recognition. When participants needed calm, autonomy, or uninterrupted focus, silence was experienced as respect. The emotional meaning of silence therefore depended on whether the absence of communication matched the user's immediate situation, expectations, and emotional state.

This theme extends the understanding of emotionally intelligent communication by showing that effective app communication is not only a matter of what is said, but also of when communication is withheld. In health and exercise applications, silence can either protect the user's autonomy or intensify frustration. The key issue is not whether the app communicates frequently, but whether its communication is sensitive to when support is needed and when restraint is more appropriate. Thus, emotionally intelligent communication requires not only appropriate messages, but also context-sensitive silence.

4.4 Theme 3: cognitive offloading and the need for transparent app logic

The third theme concerns the tension between cognitive offloading and transparent app logic. Table 7 summarises how the theme was developed from reduced expressions into three subthemes. The first subtheme concerns the relief created

when users could delegate planning or interpretation to the app. The second concerns the doubt created when recommendations, metrics, or algorithmic choices were insufficiently explained. The third shows how trust was strengthened when users could recognise the connection between their own data, the app's reasoning, and the feedback provided. Together, these subthemes show that automated app support was emotionally valuable only when it remained communicatively intelligible.

TABLE 7. Development of theme 3: cognitive offloading and the need for transparent app logic.

Reduced expressions / Codes	**Subthemes**	**Main theme**
Ready-made workout libraries reducing planning effort Delegation of routine planning to the app Automated structuring of activity or recovery decisions Reduced need for independent planning and interpretation	Relief through cognitive offloading	Cognitive offloading and the need for transparent app logic
Questioning unexplained recommendations Unclear algorithmic exercise selection Generic metrics failing sport-specific contexts Need to evaluate or compensate for hidden app logic	Algorithmic opacity creating doubt	
Transparent connection between user input and app output Trust in advice grounded in personal or biometric data Conversational AI contextualising personal data Recognition of data provenance	Trust through transparent and data-grounded logic	

In this study, cognitive offloading refers to situations in which the app reduced the user's need to plan, decide, interpret, or evaluate exercise-related activity independently. This could occur through ready-made workout options, automated recommendations, tracking summaries, recovery advice, or personalised

suggestions. The incidents in this theme were especially visible during session set-up, reflection, and next-step planning, where users relied on the app either to reduce decision-making effort or to explain what should happen next.

A clear example of cognitive offloading as emotional support appeared in H1's experience with a yoga application. The participant opened the app in a context where they wanted relaxation and relief from everyday decision-making. The ready-made content library reduced the need to plan the session independently and created an immediate sense of ease:

> H1: "Relief. In a way, just that someone else [does the planning]… wow, there's all sorts of things here… a positive surprise."

This quote shows the emotional value of not having to decide everything alone. The app's ready-made structure communicated support by taking over part of the planning burden. The participant did not need to construct a routine from scratch or decide the sequence of movements independently. Instead, the app's content created an impression that something useful had already been prepared. Emotionally, this produced relief and positive surprise. In this sense, cognitive offloading lowered the threshold for engagement: the app made it easier for the user to begin.

However, the same participant also described how this relief could quickly turn into doubt when the app's recommendation logic was unclear. After selecting a focus area, the participant questioned whether the suggested movements matched what they were looking for:

> H1: "I wonder if these are quite what I was after… a bit of a questioning surprise."

This incident illustrates the fragile nature of cognitive offloading. The app initially reduced the user's decision burden, but because it did not explain why particular movements had been selected, the participant had to begin evaluating the recommendation manually. The emotional response shifted from relief to "questioning surprise". This matters because the app was implicitly positioned as a source of guidance, yet its reasoning remained hidden. Instead of fully trusting the app's expertise, the participant had to assess whether the suggested routine made sense. Opacity in the app's recommendation logic therefore reintroduced the very cognitive work that the app had initially appeared to remove.

A similar issue appeared in H6's account of sport-specific tracking. As an experienced user, H6 did not simply accept the app's metrics as neutral or authoritative. Instead, they evaluated the logic behind the measurement system and compared it with how training load was calculated in other sports. The participant described dissatisfaction with the app's reliance on heart rate rather than more sport-specific metrics:

> H6: "In cycling, it doesn't track it based on heart rate, but rather based on power, which is maybe a slightly better metric… I'd hope to be able to track things more sport-specifically with this app."

This example shows that transparent app logic is not only about explaining recommendations verbally. When an app calculates training load, it attempts to

offload the cognitive work of evaluating physical fatigue. However, if the app's underlying logic does not appear contextually appropriate to the user, this offloading fails. For H6, the issue was not merely that the app produced data, but that the data appeared to be based on a generic model that did not fully reflect the demands of the specific activity. The participant's expertise allowed them to notice this limitation and question the credibility of the feedback. As a result, the app's authority was weakened. Rather than functioning as trusted support, the app's measurement logic became something the user had to audit.

This finding is important because it shows that experienced users may not experience automated tracking as simple support. They may interpret app-generated metrics as claims that need to be evaluated. If the app does not communicate how its calculations relate to the specific training context, users may perceive the feedback as incomplete, generic, or insufficiently context sensitive. In H6's case, the desire for more sport-specific tracking reflected a need for communication that recognised the complexity of the activity. The app's failure to do so communicated a lack of contextual sensitivity.

In contrast, H2's account showed how automated feedback could generate trust when the user understood the relationship between their own input and the app's output. The participant described how adding more data to the app made the automated comments more valuable:

> H2: "The more data you put in there, the more of these comments you can get from it… it's, like, an important thing to me."

This example demonstrates a more transparent input-output relationship. The participant understood that the app's comments were not arbitrary; they were connected to the data the user provided. This made the automated feedback feel more meaningful and trustworthy. The app's intelligence was not experienced as mysterious or imposed from above, but as something partly co-created through the user's own data practices. In this case, cognitive offloading and transparency worked together: the app helped interpret information, but the user also understood why the interpretation had become possible.

A further example of trusted algorithmic support appeared in H5's use of Oura's AI Advisor. Unlike H1, who questioned whether the app's suggested yoga movements matched their needs, H5 described the AI-based feedback as trustworthy because it was visibly grounded in data collected by the smart ring. The participant explicitly contrasted this with a broader scepticism towards generative AI:

> H5: "[…] AIs are notoriously the kind of things that, like, support [your own thoughts]… and encourage you in all your delusions, but I don't experience this as being like that at all… it however bases its answers on the data that the ring has collected from you, so it's, like, based on facts."

This example deepens the theme because it shows that transparent app logic is not only relevant to simple workout recommendations or numerical metrics. It also matters in more conversational, AI-mediated feedback. For H5, the AI Advisor was not trusted simply because it sounded supportive. It was trusted

because the participant understood that its advice was connected to their own biometric data. The app therefore allowed the user to offload part of the interpretive work of making sense of daily activity, recovery, and bodily state, without making the feedback feel arbitrary or detached from reality. In this case, the transparency of data provenance transformed the AI from a potentially unreliable "black box" into a more credible, coach-like source of reflection.

This also shows that emotionally intelligent algorithmic communication does not necessarily require the user to understand the technical model behind the system. What mattered was that the participant could recognise the basis of the advice: the app was using data that had been collected from their own body and activity patterns. This made the feedback appear personalised, data-grounded, and contextually relevant. In contrast to opaque recommendations that force users to question the app's reasoning, H5's experience suggests that cognitive offloading becomes emotionally supportive when the user can see that the system's interpretation is anchored in meaningful personal data.

Taken together, these incidents show that automated support in health and exercise applications can be emotionally valuable when it reduces decision burden, supports planning, or helps users interpret their own activity and bodily state. However, this support depends on communicative intelligibility. When users do not understand why an app recommends a particular action, selects a particular routine, or calculates a particular metric, automation may shift from support to uncertainty. By contrast, when the app's reasoning is visibly connected to user input, sport-specific logic, or biometric data, cognitive offloading can become emotionally reassuring rather than opaque.

This theme therefore highlights the importance of transparent app logic as part of emotionally intelligent communication. Users do not necessarily need full technical explanations of every calculation, but they do need enough communicative clarity to understand why the app is suggesting, measuring, or interpreting something in a particular way. In health and exercise applications, personalised communication becomes more emotionally supportive when users can recognise the reasoning behind it. Cognitive offloading creates relief when it feels competent and trustworthy, but it creates doubt when the app's logic remains hidden.

4.5 Theme 4: post-session data as bodily validation

The fourth theme concerns post-session data as bodily validation. Table 8 summarises how the theme was developed from reduced expressions into two subthemes. The first subtheme concerns how app-generated metrics, scores, badges, and summaries made subjective bodily experiences visible and interpretable. The second subtheme concerns how post-session feedback could repair earlier frustration and support longer-term progress reflection. Together, these subthemes show that post-session data was not experienced merely as information, but as communicative evidence that shaped how participants interpreted effort, success, recovery, and motivation.

TABLE 8. Development of theme 4: post-session data as bodily validation.

Reduced expressions / Codes	**Subthemes**	**Main theme**
Quantified metrics making effort visible "Black-and-white" proof of subjective exertion Visual badges or scores confirming successful activity Data confirming bodily fatigue, recovery needs, or enjoyment	Translating subjective bodily experience into visible evidence	Post-session data as bodily validation
Post-session summaries softening earlier frustration Feedback creating satisfaction after difficult interaction Cumulative data supporting long-term progress reflection Progress tracking becoming game-like or motivational	Emotional repair and cumulative progress validation	

In this study, post-session data refers to the summaries, metrics, scores, graphs, achievement markers, and progress indicators that applications present after exercise or during later reflection. The findings suggest that participants did not experience this data merely as neutral information. Instead, post-session data often translated subjective bodily experience into visible and interpretable evidence. It helped participants confirm that effort had been real, that the workout had "counted", or that the body had performed in a meaningful way.

This theme was especially visible in immediate post-session feedback and later reflection. After exercise, participants often returned to the app to interpret what had just happened physically. In these moments, the app's data did more than describe the session; it shaped how the session was emotionally understood. Positive data could generate pride, satisfaction, trust, relief, or motivation. In some cases, it also softened earlier frustration by changing the emotional meaning of the overall session. The body had already experienced the workout, but the app made that experience visible, comparable, and easier to interpret.

A clear example of this appeared in H3's account of running data. The participant described how seeing the outcome of a run created pride and motivation because the data made the physical effort visible:

> H3: "You get this feeling of pride from it… you kind of see it in black and white. It motivates you and you can challenge your mind yourself."

This quote captures the central mechanism of the theme. The participant had already completed the run, but the emotional significance of the exercise was strengthened when the app turned the experience into visible evidence. The expression "black and white" is especially important because it suggests that the data made the effort feel concrete and undeniable. The workout was no longer only a bodily memory or subjective feeling; it became something that could be seen, compared, and used as a basis for future motivation. In this sense, the app's post-session communication validated the participant's effort and transformed it into a motivational resource.

H5's experience with Oura showed a related but slightly different form of validation. In this case, the app's activity score and visual star symbol did not simply measure physical effort; they confirmed that the participant's positive bodily experience had been successful and worth recognising:

> H5: "If that activity score is… over 85, then you get a star for it… it feels like a success, that I had fun, I did well, and then I got the Oura data as proof of it on top of that."

This example shows how post-session data can validate not only effort, but also enjoyment and bodily satisfaction. The participant already felt that the activity had gone well, but the smart ring data provided additional confirmation. The phrase "as proof of it on top of that" suggests that the app's data strengthened the participant's own interpretation of the experience. The visual star operated as more than a decorative reward; it became symbolic evidence that the good bodily feeling was recognised and supported by the app's data. Here, the app communicated success by aligning its feedback with the participant's lived experience.

This alignment between bodily feeling and app-generated data also appeared in H2's account. H2 described a swimming session where earlier irritation during the workout was softened by the satisfaction created by post-session feedback:

> H2: "That is, of course, definitely the most meaningful moment… Satisfaction and a feeling of success… that minor annoyance during it kind of eased off, since I was left with such a satisfied feeling otherwise."

This incident is important because it shows that post-session data can reshape the emotional trajectory of the whole app experience. Earlier in the session, the participant had experienced friction when the smartwatch screen failed to respond to wet hands (as discussed in Theme 1). However, the post-session summary produced satisfaction and a feeling of success strong enough to reduce the emotional weight of that irritation. The feedback did not remove the earlier

problem, but it changed how the participant remembered and evaluated the session. In this way, post-session validation functioned as emotional repair. It helped the participant leave the interaction with satisfaction rather than frustration as the dominant emotional aftertaste.

H2's account also suggests that emotionally powerful feedback does not need to create an emotion from nothing. Rather, it may intensify, stabilise, or legitimise an emotion that has already begun through bodily experience. In this case, the data did not replace bodily feeling but supported it.

H4's account extended this theme from immediate post-session validation to longer-term reflection. The participant described repeatedly returning to accumulated app data to see whether their fitness was improving or declining:

> H4: "At some point, this became a proper game for me, like I'd go check this every now and then… to see whether my fitness is dropping or if it's productive training."

This example shows how bodily validation can become cumulative. For H4, the app's data did not only validate a single workout; it created an ongoing progress narrative. The participant returned to the data over time to interpret whether training was productive and whether fitness was developing in the desired direction. The phrase "proper game" suggests that the data introduced a motivational structure into reflection. Progress became something to monitor, interpret, and perhaps compete with. In this case, post-session data extended beyond the immediate feedback stage and became part of the participant's longer-term meaning-making.

However, H4's example also complicates the theme. The data was motivating partly because the participant approached it actively and analytically. The app did not simply tell the user what to feel; rather, the participant used the data as a tool for self-assessment. This connects post-session validation to the concept of expert user agency, which will be explored further in Theme 6. The validating power of the data depended not only on what the app displayed, but also on the participant's ability to actively audit and interpret the data in relation to their own training goals and bodily knowledge.

Taken together, these incidents show that post-session data functioned as more than informational feedback. It operated as emotional evidence. For H3, data made effort visible "in black and white". For H5, the activity score and star gave proof that a good bodily experience had been successful. For H2, post-session satisfaction softened earlier irritation and helped preserve a positive overall interpretation of the session. For H4, accumulated data became a longer-term game of progress and self-assessment.

This theme therefore highlights the emotional power of feedback in health and exercise applications. Post-session communication can validate bodily effort and experience, strengthen motivation, repair earlier friction, and turn isolated workouts into a broader progress narrative. At the same time, its emotional value depends on whether the data resonates with the user's own bodily experience. When app-generated feedback confirms what the user felt, it can create trust, pride, and satisfaction. When data becomes visible, understandable, and

personally meaningful, it does not merely report exercise; it helps users believe in the significance of what they have done.

4.6 Theme 5: humanised coaching and mediated authority

The fifth theme concerns moments in which app-based communication became emotionally meaningful because it resembled, simulated, or mediated coaching-like support. Table 9 summarises the development of this theme from reduced expressions into three subthemes. The first subtheme concerns applications as conduits for human coaching and accountability. The second concerns AI-based, automated, or audio-based communication that simulated relational support. The third concerns users' desire for apps to move beyond passive tracking towards more active interpretive guidance. Together, these subthemes show that app communication became more motivating when it was experienced as competent, timely, relational, and worth trusting.

TABLE 9. Development of theme 5: humanised coaching and mediated authority.

Reduced expressions / Codes	**Subthemes**	**Main theme**
App mediating communication from a human coach Coach feedback delivered through the app App functioning as a discussion or accountability platform Human guidance made visible through app infrastructure	App as a conduit for human coaching and accountability	Humanised coaching and mediated authority
Conversational AI functioning as a proxy coach Automated audio cues perceived as comforting or encouraging Milestone cues compensating for solo exercise App feedback resembling relational support	Simulated relational care through AI and automation	
Desire for proactive recommendations Frustration with passive tracking Expectation that the app should interpret patterns and suggest action Need for competent, coaching-like authority	Demand for active interpretive authority	

In this study, humanised coaching and mediated authority refer to app-based communication that made users feel guided, supported, reassured, or held accountable. This did not only involve direct human coaching. It also included AI-based advice, automated audio cues, personalised feedback, and app structures that created a coaching-like relationship between the user and the application. Authority is therefore not understood as control over the user, but as perceived competence, guidance, and interpretive support.

This theme was particularly visible during set-up, in-session interaction, post-session feedback, and next-step planning. At these stages, participants often wanted help with orientation: what to do, how the session was going, how to interpret their performance, or what to do next. The findings suggest that users did not simply want more communication. Rather, they valued communication that felt relevant, competent, appropriately timed, and trustworthy. This was especially visible in H6's coach-mediated use of TrainingPeaks and H5's use of Oura's AI Advisor.

The clearest example of mediated authority appeared in H6's use of TrainingPeaks. In this case, the app did not replace a human coach but functioned as a platform through which the coach's guidance, training plan, and feedback reached the user. The participant described using the application daily with a coach and interpreted good feedback through the app as motivational:

> H6: "When you use it daily with a coach, like as a discussion platform, then of course if you get good feedback from there… then maybe it adds to that motivation."

This example shows how app-based communication can become emotionally meaningful when it mediates a real relationship. The motivational force did not come only from the application's technical functions, but from the fact that the app carried feedback from a trusted human authority. The app therefore operated as a communicative bridge between coach and athlete. Its value was not limited to storing workouts or presenting data; it structured an ongoing coaching relationship in which feedback, accountability, and training decisions could be exchanged.

H6's description of the app as "like a discussion platform" suggests that it was experienced less as an isolated self-tracking tool and more as part of a relational training environment. The app gave the coach's communication continuity and visibility, meaning that the user did not have to interpret the training process entirely alone. In this way, the app's authority was partly relational: it came from the trusted coach whose communication the platform carried.

A related but technologically different form of humanised authority appeared in H5's experience with Oura's AI advisor. Here, the coaching relationship was not mediated through a real human coach but simulated through an AI-based conversational interface. The participant described opening the advisor, asking it to analyse their activity and daily data, and receiving feedback that felt useful and responsive:

> H5: "It answered those needs of mine… they weren't really such deep needs, but I wanted to chat a bit, look at the data, to see how that kind of a coach would evaluate my data."

This quote shows that users may knowingly engage with AI as a proxy coach. H5 did not appear to confuse the advisor with a human coach; rather, the participant treated it as "that kind of a coach", a specific form of digital authority capable of evaluating data and offering reflective feedback. The emotional need was modestly framed by the participant as "not really such deep needs", yet the account still reveals a desire for dialogue, interpretation, and external perspective. The

user did not only want raw data. They wanted the data to be interpreted by something that could respond in a coaching-like manner.

This finding extends the theme by showing that humanised coaching does not require a fully human relationship to become meaningful. In H5's case, the AI advisor became valuable because it transformed self-tracking data into conversational feedback—a process made trustworthy by the transparent data provenance discussed in Theme 3. The participant could decide when to engage with it, which also made the interaction feel appropriately timed and self-directed. The coaching authority was therefore not imposed through intrusive prompts, but accessed voluntarily when the participant wanted support, interpretation, or reassurance. This made the AI advisor feel less like a generic notification system and more like a responsive source of communicative support.

The emotional effect of this advisor was also connected to behaviour. H5 described feeling satisfied after receiving feedback that recognised activity and reminded them about recovery. The app's advice contributed to the participant's decision to take a rest day. This suggests that coaching-like communication can shape engagement not only by increasing activity, but also by legitimising recovery. In this sense, emotionally intelligent communication may support sustainable engagement by helping users interpret rest as a valid next step rather than as failure or disengagement.

A smaller but important example of humanised communication appeared in H3's experience with automated audio cues during running. In this case, the app was not providing human coaching or AI-based conversation. It was simply giving automated information during the workout. However, because the cue arrived during physical effort and provided timely reassurance, the participant described it in strongly humanised terms:

> H3: "There's always this relieving feeling associated with it, that you always get some information… it's a bit like a comforting voice… it's encouraging."

This example shows that even basic automated communication can become emotionally meaningful when timing, format, and tone fit the user's situation. The audio cue was not merely informational. During a run, when the user was physically loaded and perhaps uncertain about their performance, receiving information created relief. The participant's description of the cue as "a comforting voice" is especially significant because it anthropomorphises a simple automated feature. The app did not need to provide complex coaching to feel supportive; it needed to communicate at the right moment and in a form that reduced uncertainty and encouraged continuation.

H3's account also shows that humanised communication can emerge through predictability. The user knew that information would arrive, and this expectation itself appeared to provide reassurance. The cue functioned as a small coaching presence within the run: it marked progress, confirmed that the app was still tracking, and helped the user continue. In this way, mediated authority existed not as expertise or analysis, but as a reliable voice that accompanied the user during effort.

However, the theme also contained a contrasting finding: participants noticed when apps failed to take on a sufficiently active coaching role. This desire for coaching-like guidance also appeared outside explicitly coaching-based apps, as H4's account of Hevy illustrates. The participant used the app for workout logging, but described a desire for more coach-like feedback based on the data already stored in the system:

> H4: "It could be that you'd get some sort of a more coaching-like approach into it… like if it notices, for example, that you've been doing the same weights and reps have gone up, then like 'increase the weight'."

This quote shows that passive tracking may become insufficient when the user expects the app to interpret patterns and provide actionable guidance. H4 was not asking for vague encouragement, but for a specific form of mediated authority: the app should notice progression and translate it into a clear recommendation. The blunt example, "increase the weight", captures the kind of external push the participant felt was missing. The app had access to meaningful data, but it did not transform that data into coaching communication.

This incident is important because it reveals the boundary between recording and coaching. A logbook may support self-monitoring, but it does not necessarily create the feeling of being guided. For an experienced user, the absence of coaching-like interpretation can make the app feel passive. The user is left to perform the analytical and motivational work themselves, even though the app contains data that could support more intelligent feedback. In this sense, the lack of mediated authority resembles the silence discussed in Theme 2: the app records but does not advise.

Taken together, these examples show that humanised coaching in health and exercise applications can take several forms. It may involve a real coach communicating through the app, an AI advisor simulating reflective dialogue, automated audio cues functioning as a comforting presence, or the user's desire for more active guidance from a passive tracking tool. Across these cases, the emotional value of communication depended on whether the app helped users feel supported, guided, reassured, or accountable.

This theme also complicates the idea that app-based communication is only about personalisation or feedback. What mattered was not simply that the app provided information, but whether that information carried authority and relational meaning. When communication felt coach-like, it helped users interpret effort, make decisions, and sustain motivation. When this authority was absent, users noticed the gap and sometimes wished the app would move beyond passive recording towards more active guidance. Thus, emotionally intelligent app communication may require not only accurate data and well-timed messages, but also a sense of competent, context-sensitive authority that users can trust.

4.7 Theme 6: expert user agency: auditing and emotional distancing

The sixth theme concerns expert user agency, particularly the ways in which experienced users evaluated, questioned, limited, or adapted app-generated communication. Table 10 summarises how the theme was developed from reduced expressions into three subthemes. The first subtheme concerns emotional distancing and self-regulation, where users consciously limited the affective power of app metrics. The second subtheme concerns expert auditing and embodied judgement, where users evaluated app communication against their own training knowledge and bodily experience. The third subtheme concerns the protection of data integrity and the limiting of app authority, where users controlled how far the app was allowed to guide, interpret, or define their activity. Together, these subthemes show that app-based communication did not operate on passive users but was actively negotiated through users' own expertise and sense-making.

TABLE 10. Development of theme 6: expert user agency.

Reduced expressions / Codes	**Subthemes**	**Main theme**
Consciously limiting the emotional weight of app metrics Refusing to let numbers determine mood Decoupling physical performance from app evaluation Treating app feedback as informative but not emotionally decisive	Emotional distancing and self-regulation	Expert user agency: auditing and emotional distancing
Auditing generic calculations against sport-specific knowledge Questioning app metrics through training expertise Prioritising bodily safety over digital instructions Adapting or resisting app guidance based on embodied judgement	Expert auditing and embodied judgement	
Rejecting automated convenience to protect data accuracy Manually correcting or controlling data inputs Functionally downgrading the app to a recording tool Using the app selectively within self-defined limits	Protecting data integrity and limiting app authority	

In this thesis, expert user agency refers to the ways in which experienced users did not passively accept app-based communication, data, feedback, or algorithmic recommendations as authoritative. "Expert" refers primarily to users' exercise-related knowledge, bodily awareness, and familiarity with app-supported routines, rather than formal technical expertise. Participants evaluated app communication against their own bodily knowledge, training experience,

expectations, and goals. This involved auditing app-generated information, questioning whether metrics were meaningful, selectively trusting feedback, and emotionally distancing themselves from data when necessary.

This theme was particularly visible in post-session feedback and reflection. At these stages, apps often presented users with data, summaries, scores, or interpretations that could influence how users evaluated their own effort and progress. However, the findings suggest that experienced users did not always allow the app to define the meaning of their performance. Rather than treating app-generated data as a final judgement, they often interpreted it as one source of information among others. This theme therefore complicates the previous findings by showing that app communication can shape emotional experience, but users also actively regulate how much authority the app is allowed to have.

The clearest example of emotional distancing appeared in H4's relationship with fitness data. The participant described looking at metrics but intentionally refusing to give them too much emotional weight. Instead of allowing the numbers to determine their mood, the participant framed them as something to be observed without becoming personally destabilised by them:

> H4: "But maybe I've just gotten used to them myself, so I don't give those numbers an awful lot of weight… they don't affect my whole rest of the day."

This quote shows that emotional engagement with app data is not automatic. Although the app produced metrics that could potentially invite comparison, self-evaluation, or emotional reaction, the participant actively limited their affective significance. The data remained visible and available, but it was not allowed to dominate the participant's emotional state after the workout. This is important because it shows a form of self-regulation that sits between trust and rejection. H4 did not dismiss the app entirely, but neither did they treat its numbers as emotionally decisive.

This form of distancing can be understood as a protective strategy. In health and exercise applications, metrics may easily become emotionally loaded because they appear to quantify effort, progress, discipline, or success. For an experienced user, however, the meaning of a workout may not depend solely on what the app displays. H4's account suggests that expertise can create emotional boundaries around data. The user can see the numbers, understand them, and still decide not to let them define the rest of the day. In this sense, emotional distancing functioned as a form of agency.

H6's account showed a different form of expert agency: auditing the app's measurement logic. As discussed in Theme 3 regarding transparent app logic, H6 did not simply accept the application's metrics as neutral. Instead, the participant actively evaluated whether the app's logic matched the specific demands of the activity being measured. In discussing sport-specific tracking, H6 compared heart-rate-based measurement with power-based measurement:

> H6: "In cycling, it doesn't track it based on heart rate, but rather based on power, which is maybe a slightly better metric… I'd hope to be able to track things more sport-specifically with this app."

This example shows that expert users may interpret app-generated data as a claim that can be questioned. The app's communication was not only the number or metric itself, but the underlying logic that made the number meaningful. For H6, a generic or insufficiently sport-specific measurement model weakened the authority of the feedback. The participant's training knowledge allowed them to recognise that different sports require different ways of evaluating load and performance. As a result, the app was not treated as a complete authority, but as a system whose assumptions had to be assessed.

This kind of auditing is analytically important because it shows that transparency and expertise are connected. A less experienced user might accept a metric at face value, whereas an expert user may ask whether the metric is appropriate for the context. H6's response was not emotional disengagement from data as such. Rather, it was a demand for more precise and context-sensitive communication. The app remained useful, but its authority was conditional. It needed to demonstrate that its logic was appropriate to the activity before the user could fully trust its interpretation.

H5's relationship with Oura data illustrated a more calibrated form of trust. The participant valued the app's information but did not position it above their own bodily awareness. Instead, the app functioned as an additional source of knowledge that complemented what the participant already felt and understood about their body:

> H5: "I feel that I can already understand what my body is telling me without relying on the data. Still, the data gives me something extra to reflect on and helps satisfy my curiosity."

This quote shows selective trust rather than dependence. H5 did not reject the app's data; in fact, the participant described it as satisfying a desire for knowledge. However, the data did not replace embodied self-knowledge. The participant positioned their own bodily interpretation as primary and the app as supplementary. This matters because it challenges the idea that health and exercise applications simply tell users what their bodies mean. For H5, the app contributed to interpretation, but it did not own the interpretation.

The emotional significance of this calibrated trust lies in the participant's ability to keep the app in proportion. The app could support curiosity, reflection, and self-understanding, but it was not treated as the only valid source of truth. This differs from a passive user relationship in which app-generated scores or readiness indicators might directly determine mood or decisions. H5's account shows that expert users may engage deeply with data while still maintaining interpretive independence.

A related form of agency appeared again in H4's description of Hevy. The participant described the app primarily as a recording tool rather than as a complete workout planning or coaching system:

> H4: "Yeah, for me it is just a recording tool, and not exactly that kind of workout building or workout programming… so that is maybe the thing for me, why I surely get that annoyance with it…"

This quote shows functional downgrading. H4 had assigned the app a specific role: it was useful for recording workouts, but not for broader programming or coaching. This distinction mattered because annoyance emerged when the app's limitations became visible in relation to what the participant might otherwise want from a training tool. The participant was not confused about the app's role; instead, they had clearly defined what the app was and was not allowed to do. This is another form of user agency. The user limits the app's authority by placing it within a narrower functional category.

Functional downgrading is important because it shows that dissatisfaction does not always lead to abandonment. A user may continue using an application while mentally reducing its role. In H4's case, the app remained useful, but only within a bounded purpose. It did not become a trusted coach, motivational authority, or intelligent planner. This finding connects to the broader pattern of expert users treating apps as tools rather than as unquestioned authorities. In this case, the emotional relationship with the app appeared more stable when the user knew what the app was useful for and where its limits began.

H1's account offered a brief but important embodied example of expert user agency. A further example can be seen in the yoga incident discussed in Theme 2, where the participant encountered a sequence that felt too demanding and insufficiently guided. Rather than simply following the app's structure, the participant adjusted the experience by adding intermediate movements and questioning the safety of the routine:

> H1: "I maybe didn't really find a continuum here… I did my own intermediate movements… It felt like it wasn't even safe to do these, because they kind of went off the deep end."

Although this incident also relates to missing guidance and silence, it adds another dimension to the present theme. H1 did not passively obey the app's routine. The participant used embodied judgement—their own sense of bodily safety and suitability—to assess whether the sequence felt safe and then modified the practice accordingly. This shows that user agency may also appear during the activity itself, not only during post-session reflection. When app communication does not align with bodily experience, the user may protect themselves by resisting or adapting the app's instructions.

Taken together, these examples show that experienced users actively negotiated the authority of health and exercise applications. They did not simply receive data, feedback, routines, or recommendations as objective truth. Instead, they compared app communication with bodily knowledge, training expertise, situational context, and personal goals. Sometimes this meant emotionally distancing themselves from metrics. Sometimes it meant auditing app logic. Sometimes it meant using the app as an additional source of information while retaining self-trust. In other cases, it meant downgrading the app to a narrower functional role or modifying the activity when the app's guidance felt insufficient.

This theme is important because it prevents a technologically deterministic interpretation of the findings. The previous themes show that app communication can shape emotional experience, motivation, trust, and engagement. However, Theme 6 shows that users also shape the emotional power of app

communication. Especially among experienced users, emotionally intelligent communication is not only about making app feedback more persuasive or personalised. It is also about respecting the user's existing expertise, bodily knowledge, and capacity for judgement.

For health and exercise applications, this suggests that emotionally intelligent communication should not assume that the app must always lead. In some contexts, the most supportive communication may be communication that explains its logic, offers guidance without overclaiming authority, and allows users to interpret data in relation to their own experience. Expert users may value apps most when they function as trustworthy tools rather than controlling voices. Emotional engagement, therefore, is not created by making the app more dominant, but by designing communication that users can trust, question, adapt, and integrate into their own sense-making.

4.8 From critical incidents to final themes

The findings presented above were developed through a combination of Critical Incident Technique-informed analysis and thematic analysis. The Critical Incident Technique supported the identification of concrete app-use moments that participants experienced as emotionally significant, while thematic analysis was used to interpret the broader communicative patterns connecting these incidents.

Table 11 shows how reduced critical incident types and codes were organised into subtheme-level incident clusters and, finally, into the six final themes. The purpose of the table is not to quantify the findings, but to show how the themes were grounded in specific forms of app-related experience and how individual incidents contributed to broader communicative patterns.

TABLE 11. Summary of critical incident types, incident clusters, and final themes.

Critical incident types / codes	**Subtheme / incident cluster**	**Main theme**
Unresponsive controls Forced manual interaction Rigid or premature app timing Inability to correct flawed data Missing scalable alternatives	System friction and structural rigidity	The meta-communicative role of format: system friction versus structural support

Structurally supportive access to core actions	Format as structural and situational support	
Missing transition or safety guidance Uncommunicated tracking failures Absence of expected encouragement Passive tracking without interpretive support	Silence as abandonment or lack of support	The contextual duality of silence: respect and abandonment
Absence of commercial interruptions Non-intrusive app environments	Silence as respect for autonomy and focus	
Ready-made planning Delegated routine creation Automated activity or recovery guidance	Relief through cognitive offloading	Cognitive offloading and the need for transparent app logic
Unclear recommendations Opaque algorithmic choices Generic metrics	Algorithmic opacity creating doubt	
Trust in data-grounded advice	Trust in transparent and data-grounded logic	
Post-workout summaries Visible proof of effort Visual scores or badges Data confirming fatigue, recovery, or enjoyment	Translating bodily experience into visible evidence	Post-session data as bodily validation

Feedback softening earlier frustration Cumulative progress tracking	Emotional repair and cumulative progress validation	
Coach-mediated feedback App-based accountability	App as a conduit for human coaching and accountability	Humanised coaching and mediated authority
Conversational AI advice Automated audio cues Milestone encouragement	Simulated relational care through AI and automation	
Frustration with passive tracking Desire for proactive coaching-like recommendations	Demand for active interpretive authority	
Emotional distancing from metrics Refusal to let numbers determine mood	Emotional distancing and self-regulation	Expert user agency: auditing and emotional distancing
Auditing app logic through training expertise Prioritising bodily judgement	Expert auditing and embodied judgement	
Manual correction of data Functional downgrading of the app Selective use within self-defined limits	Protecting data integrity and limiting app authority	

The incident types summarised in Table 11 show that emotionally significant app experiences were not limited to obvious moments such as notifications or explicit feedback. Instead, participants described emotional significance in a wider range of app-use situations, including moments where the app failed to respond, remained silent, structured activity automatically, displayed bodily data, mediated

coaching-like authority, or was actively questioned by the user. This supports the broader communicative interpretation of the findings: app-based communication was experienced not only through message content, but also through structure, absence, automation, feedback, relational cues, and users' own interpretive agency.

This incident-based summary also shows how individual incidents, subtheme-level clusters, and the six final themes were connected. Some incident types were situated and momentary, such as unresponsive controls, missing transition guidance, uncommunicated tracking failures, or automated audio cues during exercise. Others were broader and more cumulative, such as cumulative progress tracking, emotional distancing from metrics, functional downgrading of the app, or the user's gradual calibration of trust in app-generated data. Together, these incident types show how both micro-level app interactions and macro-level user–app relationship patterns contributed to the development of the final themes.

5 DISCUSSION AND CONCLUSIONS

This final chapter discusses the findings of the study in relation to the research questions, theoretical framework, and previous research. The purpose of the chapter is to move from the presentation of results to their broader interpretation: what the findings suggest about app-based communication, emotionally engaging user journeys, and emotionally intelligent communication in health and exercise applications. As established in the introduction, health and exercise applications are understood here as applications used by participants in relation to exercise, training, recovery, bodily self-monitoring, or exercise-related reflection.

The chapter first summarises the main findings and answers the research questions. It then discusses how the findings relate to earlier literature on customer journeys, emotional journeys, critical incidents, emotionally intelligent communication, and engagement in digital service environments. After this, the chapter presents the theoretical and practical implications of the study, with particular attention to what the findings suggest for communication professionals and app developers. Finally, the chapter evaluates the study, proposes directions for future research, and closes with a concluding synthesis of the study's contribution.

5.1 Summary of the main findings and answers to the research questions

This study examined how users perceived app-based communication as shaping and supporting emotionally engaging user journeys in health and exercise applications. The findings suggest that app-based communication is not limited to explicit messages, notifications, feedback, or coaching prompts. Rather, participants interpreted the broader app experience communicatively: how the app was structured, when it spoke, when it remained silent, how it presented data, how clearly it explained its logic, and how much authority it assumed in relation to the user's own bodily knowledge.

The central finding of the study is that emotionally engaging user journeys are shaped through the relationship between app communication, embodied experience, and user interpretation. Communication was perceived as supportive when it helped users feel competent, understood, respected, and able to make sense of their own activity. By contrast, communication was perceived as weakening the user–app relationship when it created friction, opacity, pressure, abandonment, or unnecessary authority. Across the six themes, app communication was therefore experienced not only as content, but as the overall way in which the application structured, timed, withheld, personalised, justified, and mediated interaction.

In relation to RQ1, the findings show that participants experienced a range of emotional responses at different stages of app interaction. During session set-up and in-session interaction, emotions such as relief, annoyance, frustration, uncertainty, trust, and serenity were shaped by how easily users could begin or continue an activity, whether the app provided sufficient guidance, and whether its format supported the physical context of use. In the immediate post-session feedback and reflection stages, emotions such as pride, joy, trust, disappointment, doubt, and emotional distancing were shaped by how the app presented data, explained feedback, or allowed users to interpret their performance. These emotional responses were not evenly distributed across the whole journey. Instead, they were especially visible around set-up, in-session interaction, post-session feedback, reflection, and next-step planning.

In relation to RQ2, the findings suggest that participants perceived emotional responses as connected to motivation and engagement, particularly through whether the app felt supportive, trustworthy, frustrating, intrusive, validating, or worth returning to. Positive emotional responses were often described in relation to moments where the app reduced cognitive effort, validated physical effort, provided meaningful feedback, or supported the user's autonomy. Negative emotional responses were described when communication was poorly timed, missing, overly rigid, insufficiently transparent, or misaligned with the user's embodied situation. However, the relationship between emotion and engagement was not linear. Frustration did not always lead to abandonment, particularly when users could downgrade the app's role, compensate through their own expertise, or continue using the application for a narrower purpose.

In relation to RQ3, the findings indicate that emotionally intelligent communication in health and exercise applications requires more than encouraging language or personalised prompts. It requires communicative sensitivity to context, timing, user autonomy, bodily experience, and interpretive agency. The findings suggest that effective app communication should reduce unnecessary friction, provide guidance when users need support, remain silent when interruption would be intrusive, make app logic sufficiently understandable, validate effort without overclaiming authority, and respect users' own expertise. In this sense, emotionally intelligent communication is not only about making the app more expressive, but about making it more responsive, intelligible, and appropriately restrained.

For corporate communication, these findings suggest that the emotional quality of digital service encounters depends not only on message content, but

on the broader communicative intelligence of the service. In health and exercise applications, this intelligence lies in knowing when to guide, when to explain, when to validate, when to remain silent, and when to leave interpretive authority with the user. The application can therefore be understood as an ongoing communicative touchpoint through which users evaluate not only the app's functionality, but also its responsiveness, competence, trustworthiness, and respect for the user.

Overall, the findings address the broader research aim by suggesting that communication may shape and support emotionally engaging user journeys when users perceive it as supportive, intelligible, respectful, and responsive across different stages of app use. At the same time, participants described communication as potentially weakening engagement when it created friction, opacity, pressure, abandonment, or emotional overload. Because this study is based on qualitative interview accounts, the findings do not demonstrate actual long-term retention or behavioural change. Instead, they show how participants perceived app-based communication as shaping motivation, engagement, avoidance, trust, and willingness to continue use. These findings provide the basis for the following discussion of the study's theoretical contributions, practical implications, and evaluation.

5.2 Theoretical contributions and relation to previous research

This section discusses the findings in relation to previous research and outlines the theoretical contributions of the thesis. The study contributes to the understanding of app-based communication, emotional journeys, and customer journey research in four main ways. First, it broadens the concept of communication in health and exercise applications by showing that communication is not limited to explicit messages, notifications, or feedback texts. Second, it contributes to emotional journey research by showing how emotional meaning is shaped through the interaction between app structures, communicative cues, bodily experience, and user interpretation. Third, it extends the concept of emotionally intelligent communication by emphasising not only responsiveness and empathy, but also restraint, intelligibility, and respect for user agency. Fourth, it highlights the role of user agency, suggesting that emotional engagement can be actively co-constructed between the app and the user rather than passively received.

The first contribution concerns the communicative nature of app interaction. The findings support earlier customer journey research that conceptualises customer experience as multidimensional and dynamic rather than as a single service encounter. Lemon and Verhoef (2016) argue that customer experience unfolds across multiple touchpoints and includes cognitive, emotional, behavioural, sensorial, and social dimensions. Similarly, Følstad and Kvale (2018) emphasise the experiential and customer-centred nature of journeys. The findings of this thesis align with these perspectives by showing that participants' emotional experiences were shaped across several stages of app use, including session set-up, in-session interaction, post-session feedback, reflection, and next-step planning.

However, the findings also extend customer journey literature by foregrounding communication as a central mechanism through which emotional meaning is produced. Previous research has acknowledged the importance of touchpoints, interaction quality, and emotional responses across the journey (Lemon & Verhoef, 2016; Mele et al., 2021; Voorhees et al., 2017). This study adds to this by showing that, in health and exercise applications, communicative meaning is not limited to messages, notifications, or feedback texts. Participants also interpreted format, silence, data visibility, the transparency of app logic, and mediated coaching structures as communicative. Interpreted through Watzlawick et al.'s (1967) interactional view of communication, this supports the idea that communication includes not only what is explicitly said, but also what is withheld, implied, structured, or made possible through interaction.

This broader view of communication is especially important in digital app environments, where many emotionally significant interactions occur through subtle or indirect cues. For example, an unresponsive interface during swimming was not merely a usability issue, but an emotionally disruptive moment because it occurred during physical activity and interfered with the participant's attempt to record a meaningful workout. Similarly, silence could communicate respect when it protected the user's focus, but abandonment when guidance or warning was needed. App format could communicate support when interaction was smooth and accessible, but rigidity when users were unable to correct mistakes or continue an activity without disruption. The thesis therefore contributes a broader conceptualisation of app-based communication as a system of explicit and implicit cues embedded in the structure of interaction.

From a corporate communication perspective, this positions the health or exercise application as an ongoing organisational touchpoint through which the user relationship is maintained, weakened, or repaired. Rather than functioning only as a technical service interface, the application becomes part of the organisation's relationship-building with users: it communicates reliability, responsiveness, care, competence, or neglect through repeated interactions. This links the findings to relationship-oriented understandings of corporate communication and relationship management, where trust, satisfaction, commitment, and perceptions of responsiveness are shaped through continuous interaction rather than isolated messages (Ledingham, 2003; Ledingham & Bruning, 1998). It also connects with Luoma-aho's (2015) understanding of stakeholder engagement as involving cognitive, emotional, and behavioural dimensions, suggesting that app-based interactions may matter not only as functional service encounters, but also as moments through which users evaluate their relationship with the organisation.

This finding also extends recent corporate communication and public relations research on digitalisation and mediated organisation–stakeholder relationships. Brockhaus et al. (2022) emphasise that communication technology and digital infrastructure are increasingly embedded in how corporate communication is organised and practised, while Lock (2019) argues that digital relationships are constituted through mediated communication. This study adds a user-level perspective to the discussion by showing how organisational responsiveness, care,

competence, or neglect may be inferred through repeated app-based interactions, including feedback, silence, automation, and data presentation.

The second contribution concerns emotional journey research. The findings connect closely with the emotional journey perspective, which views customer experiences as transitory and emotionally charged rather than stable or predictable (Palmer, 2010). This study supports that view by showing how participants' emotional interpretations shifted across app-use moments. For example, irritation during a swimming session could later be softened by post-session feedback that created satisfaction and a feeling of success. This illustrates how emotional experience can change within the same journey: a negative incident does not necessarily define the entire app experience if later communication repairs, validates, or reframes the user's interpretation.

In this study, the emotional journey should therefore not be understood as a fixed sequence of emotional states attached to predefined customer journey stages. Instead, the findings suggest that emotional journeys are situationally constructed through critical interaction moments. The same communicative absence could be experienced as abandonment in one context and respect in another. Similarly, post-session data could create pride when it validated bodily effort, but doubt when its logic was unclear or misaligned with the user's experience. This contributes to emotional journey theory by showing that the emotional meaning of a touchpoint depends not only on its position in the journey, but also on its contextual fit with the user's embodied state, expectations, and interpretive situation.

The findings also support the value of the Critical Incident Technique for identifying emotionally meaningful app interactions. Edvardsson and Roos (2001, p. 253) define critical incidents as interaction incidents that customers remember as unusually positive or negative. In this study, critical incidents were not always dramatic failures or major successes. Instead, emotionally meaningful moments often appeared in small interactional details: an app failing to warn the user about tracking problems, a lock-screen widget making it easier to stop a run, an AI advisor explaining recovery through personal data, or a post-session summary confirming effort. This aligns with contemporary qualitative uses of CIT, which emphasise the participant's interpretation of why a moment mattered rather than only the observable behaviour itself (Butterfield et al., 2005; Watkins et al., 2022).

The study therefore contributes to critical incident research by showing that, in health and exercise applications, emotionally significant incidents may be embedded in micro-communications. A small moment can become critical when it intersects with bodily effort, vulnerability, motivation, or self-evaluation. This is particularly relevant in exercise-related contexts, where users may already be physically tired, self-critical, hopeful, or uncertain. A technical or communicative failure may therefore carry emotional weight beyond its functional importance. Conversely, a small supportive cue, such as audio feedback during a run or visible post-session progress, may become motivating because it arrives at a moment when the user is especially receptive to reassurance or validation.

The third contribution concerns emotionally intelligent communication. Earlier research has described emotional intelligence as the ability to recognise,

understand, and respond appropriately to emotions (Goleman, 1995, 1998; Mayer & Salovey, 1997). In organisational and communication contexts, emotionally intelligent strategies involve empathy, timing, relevance, and sensitivity to the audience's affective readiness (Hess & Bacigalupo, 2011; Sigmar et al., 2012). The findings of this thesis suggest that these principles are highly relevant to health and exercise applications. Participants valued communication that reduced friction, supported autonomy, explained feedback, validated effort, and respected their own judgement. In contrast, emotionally unintelligent communication appeared as missing guidance, poorly timed interruption, insufficiently explained app logic, or rigid system design.

However, the findings also refine the concept of emotionally intelligent communication. In app-based contexts, emotional intelligence does not necessarily mean more expressive, more encouraging, or more human-like communication. In some cases, participants valued silence, restraint, and non-interruption. This is important because emotional responsiveness in apps can easily be misunderstood as adding more motivational prompts or personalised messages. The findings suggest a more conditional view: communication is emotionally intelligent only when it fits the user's situation. Sometimes this means offering guidance or encouragement; at other times, it means stepping back and allowing the user to continue without interference.

This also resonates with Dijkmans et al.'s (2015) finding that conversational human voice can shape how publics evaluate organisations in mediated environments. However, this study extends the idea to app-based interaction by showing that perceived human presence is not always desirable in itself; its value depends on whether the communication feels timely, credible, and contextually appropriate.

This finding connects with research on barriers to emotionally intelligent communication. Santos and Ponchio (2021) argue that emotional discomfort can contribute to resistance in digital services, while Evert et al. (2023) note that digital mediation may reduce the emotional richness of communication. This study supports these concerns by showing how poorly aligned communication was associated with frustration, doubt, or disengagement. However, the findings also show that emotional barriers are not always caused by excessive communication. They can also emerge from communicative absence. Kantola and Kannasto (2026) similarly highlight how silence, delayed responses, and non-response in digital communication can become ambiguous and emotionally consequential when recipients are left to interpret the meaning of absence. In this study, this dynamic appeared when an app failed to explain, warn, guide, or acknowledge the user's situation, making absence itself communicatively significant. Thus, emotionally intelligent communication requires not only better messages, but also better judgement about when communication is needed and when it should be withheld.

The role of transparent app logic further extends previous discussions of personalisation and adaptive digital communication. This also aligns with Liu's (2021) finding that meaningful transparency in human–AI interaction can reduce uncertainty and increase trust, suggesting that explanations matter not only technically but also communicatively. Märtin et al. (2023) highlight the potential of

adaptive systems to respond dynamically to user behaviour and emotional states. The findings of this thesis support the value of adaptive and personalised communication, but they also show that personalisation alone is not sufficient. Participants trusted app-generated advice more when they could understand the basis of the recommendation or connect it to meaningful personal data. For example, trust in AI-based advice was linked to the perception that the advice was based on data collected by the device rather than on generic encouragement. Similarly, sport-specific criticism of app metrics showed that experienced users may question feedback when they do not understand or accept the logic behind it. This suggests that adaptive communication must be not only personalised, but also intelligible.

The findings also connect with research on conversational agents and communicative AI in organisation–stakeholder interactions. Syvänen and Valentini (2020) note that conversational agents are increasingly used in online stakeholder interaction, while their organisational, ethical, and stakeholder implications remain insufficiently examined in corporate communication research. Lindholm and Sihvonen (2024) similarly emphasise that communicative AI should be understood not only as a technical tool, but as an interactional participant whose human-like communication style, perceived agency, and limitations may shape users' emotions, trust, and critical evaluation. The findings support this concern: AI-based or coach-like app communication was emotionally valuable when users experienced it as data-grounded, transparent, and proportionate, but such communication also created expectations of competence and responsibility. This suggests that automated coaching should be analysed not only as a technical feature, but as a form of organisational communication that may influence trust, perceived care, and user agency.

The fourth contribution concerns user agency in app-based emotional journeys. The findings complicate assumptions about users as passive recipients of app-based communication. This aligns with computer-mediated communication research showing that engagement in AI-supported environments can emerge through interaction between human agency and machine agency rather than through one-directional technological influence (Kang & Lou, 2022). Several participants, especially experienced users, actively audited, questioned, downgraded, or emotionally distanced themselves from app feedback. This finding is important because much customer journey and app engagement literature focuses on how organisations can shape user experience through touchpoints, messages, and design. While this perspective is useful, this study shows that users also shape the meaning and emotional force of communication. App feedback does not simply act on users; users interpret it through bodily knowledge, training experience, expectations, and personal goals.

This was particularly visible in the theme of expert user agency. Some participants did not allow app metrics to determine their mood, while others treated app data as supplementary to bodily awareness rather than as a replacement for it. Another participant evaluated whether the app's measurement logic was sufficiently sport specific. These findings suggest that emotionally engaging communication should not aim to maximise user dependence on the app. Instead, it should support informed interpretation, self-trust, and appropriate user control.

This extends emotionally intelligent communication theory by highlighting agency as a central part of emotional safety and trust. Users may experience app communication as more supportive when it respects their capacity to judge, question, and contextualise feedback.

Taken together, these contributions suggest that emotionally engaging app journeys should not be understood as products of isolated features or single messages. They emerge from a communicative ecology in which format, silence, feedback, app logic, data presentation, coaching authority, and user agency interact across time. The thesis therefore contributes a communication-centred understanding of emotional journeys in health and exercise applications. It shows that emotionally intelligent app communication is not only about delivering the right message, but about designing interactional conditions in which users can feel supported, understood, competent, and respected.

In theoretical terms, the study bridges customer journey research, emotional journey theory, emotional intelligence, critical incident thinking, and app-based communication by showing how these frameworks can be integrated in the analysis of digital service experiences. The customer journey provides the temporal and interactional structure; the emotional journey highlights affective shifts across that structure; CIT helps identify emotionally significant moments within the journey; emotional intelligence explains the need for context-sensitive and empathetic communication; and the broad interactional view of communication reveals how meaning can be created through both presence and absence. By combining these perspectives, the thesis offers a framework for understanding app-based communication as an emotionally consequential part of digital customer experience.

5.3 Practical implications for emotionally intelligent app communication

The findings have practical implications primarily for communication professionals and organisations designing health and exercise applications. Applying these implications would, however, require collaboration with UX writers, designers, developers, and product teams. This thesis does not aim to provide a technical design model; instead, the findings suggest several principles for emotionally intelligent app communication. The purpose is not to prescribe interface design solutions, but to clarify how organisations can communicate more intelligently through digital service touchpoints. In health and exercise applications, communication strategy is not limited to campaign messages or notification copy; it is embedded in the way the service responds, guides, explains, withholds, validates, and recognises the user. This aligns with recent strategic communication thinking, which emphasises that communication in changing conditions requires flexibility, contextual judgement, continuous evaluation, and the ability to adapt rather than relying only on rigid, linear planning (Juholin & Rydenfelt, 2025). The principles below therefore concern how applications can communicate in ways that are supportive, context-sensitive, intelligible, and respectful of user agency.

First, emotionally intelligent app communication should treat format as part of communication. The findings suggest that participants did not experience format as a neutral technical background. Instead, responsiveness, editability, access to controls, interaction flow, and visual structure shaped whether the app felt supportive or obstructive. This means that communication professionals should not focus only on message wording, tone, or notification content. They should also consider whether the communicative situation allows users to respond, adjust, or act at the moment communication appears. A motivational message, feedback summary, or coaching prompt loses emotional value if the surrounding interface is difficult to use, poorly timed, or incompatible with the user's physical context.

For health and exercise applications, this is especially important because users often interact with the app while physically active, tired, sweaty, wet, distracted, or emotionally vulnerable. Communication should therefore be designed around the user's embodied situation. Controls should be easy to access during activity, feedback should be available when users are ready to interpret it, and app structures should allow correction when ordinary human mistakes occur. From a practical standpoint, emotionally intelligent communication requires coordination between UX design, interface structure, and communication strategy. The message and the medium cannot be separated.

Second, app communication should distinguish between helpful silence and harmful silence. The findings showed that silence can be interpreted either as respect or abandonment, depending on context. This has a clear practical implication: more communication is not automatically better communication. In some moments, users value non-interruption, calm, and autonomy. In other moments, silence feels like a failure to guide, warn, or recognise the user's situation. Health and exercise applications should therefore avoid both extremes: constant prompting on one side and complete communicative absence on the other.

In practice, this means that apps should communicate when communication reduces uncertainty, protects safety, supports interpretation, or prevents avoidable frustration. For example, if tracking fails during exercise, the user should be informed while corrective action is still possible. If a routine requires careful bodily positioning or progression, the app should provide sufficient guidance rather than leaving the user to guess. However, when the user is engaged in focused activity, reflection, recovery, or self-directed practice, the most emotionally intelligent choice may be restraint. Silence becomes supportive when it protects the user's attention and autonomy.

Third, applications should make automated and app-generated communication sufficiently understandable. The findings showed that users appreciated cognitive offloading when apps reduced planning burden, helped interpret data, or suggested next steps. However, automated recommendations created doubt when the app's reasoning was unclear. This suggests that emotionally intelligent personalisation requires explanation, not only prediction. Users do not necessarily need detailed technical descriptions of algorithms, but they do need enough information to understand why a recommendation, score, or feedback message is relevant.

Practically, this could involve short explanations such as "recommended based on your recent recovery trend", "suggested because your training load has increased this week", or "based on your previous running pace". Such explanations may help users connect app communication to their own data, behaviour, and bodily state. This is especially important in health and exercise applications, where algorithmic feedback may influence how users interpret their effort, recovery, progress, or readiness. When the basis of communication is visible, users may be more likely to experience automated support as competent and trustworthy rather than arbitrary or opaque.

Fourth, post-session feedback should be designed as emotional interpretation, not merely data delivery. The findings showed that post-session data often validated participants' bodily effort. Metrics, graphs, activity scores, and visual symbols helped users feel that their effort had counted and that their bodily experience had been recognised. This suggests that post-session communication should help users make emotionally meaningful sense of what they have done.

However, validation should be handled carefully. Feedback should affirm effort without exaggerating, moralising, or reducing the user's worth to numbers. In health and exercise contexts, users may already attach strong emotional meaning to performance, recovery, and progress. Therefore, post-session feedback should communicate accomplishment, learning, or continuity rather than only ranking, comparison, or optimisation. For example, instead of focusing exclusively on whether a user achieved a target, feedback could acknowledge consistency, effort, recovery, adaptation, or the value of completing a session under imperfect conditions. This may support a more emotionally sustainable relationship with exercise data.

Fifth, coaching-like communication should be used carefully and credibly. The findings showed that humanised coaching, mediated authority, and supportive audio or AI-based feedback could strengthen motivation when the communication felt competent, relevant, and appropriately timed. Users appreciated guidance when it helped them interpret performance, reduce uncertainty, or decide what to do next. However, this does not mean that all apps should become more human-like or conversational. Coaching-like communication creates expectations of competence and responsibility. If the app sounds like a coach but cannot explain its reasoning, adapt to context, or recognise user needs, the humanised tone may feel hollow or intrusive.

For practice, this means that organisations should align communicative tone with actual system capability. If an app offers simple automated feedback, it should not overclaim personal understanding. If it provides AI-based advice or coach-like recommendations, it should make clear what the advice is based on and where its limits are. Humanised communication should support trust, not simulate intimacy without substance. In health and exercise applications, users may be particularly sensitive to advice that concerns bodily performance, fatigue, recovery, or self-discipline. Coaching-like communication should therefore be respectful, transparent, and proportionate.

Sixth, emotionally intelligent communication should respect user agency and expertise. The findings showed that experienced users often audited app-generated feedback, questioned metrics, downgraded the app's role, or relied on

their own bodily knowledge. This suggests that app communication should not assume that users want the application to be the final authority. Especially in health and exercise contexts, users may have different levels of experience, different training goals, and different relationships with data. Some may want detailed guidance; others may want flexible support, optional explanations, or space for independent interpretation.

A practical implication is that app communication should offer adjustable depth. Novice users may benefit from more explanatory guidance, while experienced users may value customisation, transparency, and the ability to interrogate or modify recommendations. Rather than presenting feedback as unquestionable truth, apps can communicate in ways that invite reflection. For example, they might frame insights as suggestions, patterns, or possible interpretations rather than absolute judgements. This may protect user autonomy and reduce the risk that app-generated communication becomes emotionally controlling or overly authoritative.

Overall, these practical implications suggest that emotionally intelligent app communication is not simply warmer wording or more frequent encouragement. It is a broader communication strategy that integrates message content, timing, format, explanation, silence, feedback, and user control. In health and exercise applications, emotionally intelligent communication should help users feel supported without being pressured, guided without being controlled, and validated without being reduced to data.

For communication professionals, and for UX writers working with them, the central implication is that app-based communication should be designed as part of the whole user journey rather than as a set of isolated messages. Notifications, feedback summaries, coaching prompts, interface flows, silence, and data visualisations all contribute to how users interpret the app relationship. The role of communication professionals is therefore not only to write better messages, but to help ensure that the app's communicative behaviour is coherent, ethical, and emotionally attuned.

In practical terms, this study suggests the following principles for emotionally intelligent app communication: reduce unnecessary friction; provide guidance when users need support; avoid interruptions when users need focus; explain the basis of recommendations; validate effort in emotionally sustainable ways; use coaching-like communication only when it is credible; and respect users' ability to interpret, question, and adapt app feedback. These principles do not form a universal checklist, but they offer a communication-centred direction for designing health and exercise applications that may support emotionally engaging and ethically responsible user journeys.

5.4 Evaluation and limitations of the study

This section evaluates the scope, strengths, trustworthiness, and limitations of the study. As a qualitative and interpretive study, the aim was not to produce statistically generalisable findings, but to develop an in-depth understanding of

how participants made sense of emotionally significant app-based communication. The quality of the study is therefore considered in terms of trustworthiness, transparency, reflexivity, and transferability rather than statistical reliability or representativeness (Lincoln & Guba, 1985; Nowell et al., 2017).

A key strength of the study lies in its ability to connect concrete app-use incidents with broader communicative and emotional mechanisms. By combining stimulated recall interviews, Critical Incident Technique-informed questioning, Plutchik's Wheel of Emotions, and reflexive thematic analysis, the study was able to examine not only what participants felt, but also how they interpreted the communicative conditions that shaped those feelings. This approach made it possible to identify subtle but meaningful dynamics, such as silence functioning either as respect or abandonment, post-session data operating as bodily validation, and experienced users emotionally distancing themselves from app-generated metrics.

Another strength of the study is its broad understanding of communication. Rather than limiting communication to verbal messages, notifications, or feedback texts, the analysis treated format, timing, absence, personalisation logic, data presentation, and mediated coaching structures as communicative when participants interpreted them as meaningful. This supported the corporate communication focus of the thesis by showing that digital service communication is not restricted to message design but is embedded in the wider structure of interaction between the user, the application, and the organisation behind it.

Trustworthiness was supported through methodological transparency and systematic documentation of the analytical process. Credibility was strengthened through stimulated recall, app-based reflection, and Critical Incident Technique-informed questioning, which helped anchor participants' accounts in concrete app-related experiences rather than general opinions about health and exercise applications. Dependability was supported through a semi-structured interview guide and a systematic analysis process. Although the interviews allowed flexibility and follow-up questions, they followed the same overall logic, moving from app-use context to customer journey reflection, emotion reflection, and discussion of emotionally significant incidents.

The analysis was organised through a structured incident-based workbook that documented participant codes, app contexts, journey phases, communicative elements, emotion labels, behavioural implications, and analyst memos. This created a traceable path from the original interview material to the final themes, which is important for trustworthiness in thematic analysis (Nowell et al., 2017). Confirmability was strengthened by maintaining a distinction between participants' own expressions and the researcher's interpretations. Participants' original Finnish expressions were retained during analysis, and Plutchik-based categories were used as supportive labels rather than replacements for participants' own wording. Interpretive decisions were documented through analyst memos, and emerging themes were repeatedly checked against the original transcripts, coded incidents, research questions, and selected quotations.

Reflexivity was especially important because the study approached communication broadly. As the researcher, I actively shaped the study through the theoretical framework, interview design, follow-up questions, coding decisions,

and theme development. My own familiarity with health and exercise applications supported sensitivity to app-related routines and terminology, but it also required caution to avoid projecting my own assumptions onto participants' accounts. To reduce this risk, interpretations were repeatedly checked against concrete incidents, participant quotations, the analysis workbook, and the research questions. The themes should therefore be understood as the result of active interpretation rather than as findings that simply emerged from the data without researcher involvement (Braun & Clarke, 2006, 2019). AI tools were used only as supportive aids for organisation, comparison, and reflection, while final analytical responsibility remained with the researcher.

Transferability was addressed by providing contextual detail about participants, app-use situations, applications discussed, and methodological choices. The study does not aim to produce statistically generalisable conclusions about all health and exercise app users. Instead, it offers analytically transferable insights into how app-based communication may shape emotional experience in comparable digital service contexts. In line with qualitative research principles, transferability depends on the contextual detail provided rather than statistical representativeness (Lincoln & Guba, 1985).

The findings should also be interpreted within the limitations of the study. First, the sample consisted of six Finnish-speaking young adults, several of whom had relatively high levels of exercise experience. The findings are therefore not intended to represent all users of health and exercise applications. The participant group may also have shaped the nature of the findings, as experienced exercisers may interpret app-based feedback, data, guidance, and app logic differently from novice users or users with more ambivalent relationships to exercise.

Second, the data were based on participants' recalled and narrated accounts rather than direct observation, app analytics, or physiological emotion measures. Stimulated recall and app-based reflection helped anchor the interviews in concrete experiences, but retrospective accounts may still be shaped by memory, post-hoc interpretation, social desirability, or participants' desire to make their experiences coherent. The findings should therefore be read as interpretations of perceived emotional and communicative experience, not as direct measurements of emotion or behaviour.

Third, the study included different types of health and exercise applications, including tracking, yoga, running, recovery, workout logging, and coach-mediated applications. This variation enriched the data by allowing the analysis to identify communicative patterns across app contexts. However, it also limits the possibility of making detailed claims about one specific application, platform, or feature. The findings should therefore be understood as patterns across app-based communication experiences rather than evaluations of individual applications.

Fourth, the use of Plutchik's Wheel of Emotions supported emotional reflection but may also have shaped how participants labelled their emotions. Although participants were encouraged to use their own words alongside Plutchik-based categories, any predefined emotion model simplifies complex, mixed, and context-specific emotional experiences. In addition, the interviews were

conducted and analysed in Finnish but reported in English. Translation was necessary for thesis reporting, but emotional nuance, idiomatic expression, and tone may shift when interview excerpts are translated (van Nes et al., 2010). To mitigate this limitation, the translated excerpts were checked against the original Finnish material, with particular attention to preserving the emotional meaning and interpretive context of participants' expressions.

Finally, the study focused on participants' perceptions of motivation, engagement, avoidance, trust, and willingness to continue using applications. It did not measure long-term retention, app usage behaviour, or behavioural change over time. Therefore, claims about engagement and retention should be understood as perception-based and interpretive rather than as evidence of actual long-term behavioural outcomes.

5.5 Future research

Future research could extend this study in several ways. First, larger and more diverse samples could examine whether users with different levels of exercise experience, age, cultural background, health status, or relationships with physical activity interpret app communication differently. This would be especially valuable for studying users with lower exercise confidence, chronic illness, disability, injury recovery, or more vulnerable relationships to fitness and bodily self-monitoring.

Second, future research could focus on narrower app categories, such as recovery trackers, running applications, AI coaching tools, strength training logs, or yoga applications. A narrower design would allow more precise comparison of communicative features within a specific app context. AI-mediated communication would be especially important to examine further, as conversational AI and automated coaching tools raise questions about authority, credibility, emotional tone, transparency, and ethical boundaries.

This direction is supported by recent computer-mediated communication research showing that AI chatbot interactions can involve emotionally complex experiences, including attachment, frustration, uncertainty, and technical breakdowns (Li & Zhang, 2024). In health and exercise applications, similar questions may become increasingly relevant as conversational AI tools are used to interpret bodily data, recovery, motivation, and progress.

Third, longitudinal research could examine how repeated exposure to app communication shapes trust, motivation, irritation, dependence, disengagement, or sustainable exercise habits over time. This would help clarify how emotionally significant app experiences accumulate across longer-term user journeys, rather than only being interpreted through recalled individual incidents.

Fourth, future studies could combine qualitative interviews with diary methods, screen recordings, experience sampling, in-field observation, or app-use data. Experience sampling could capture users' emotional responses closer to the moment of interaction, while in-field observation could allow researchers to examine how app-based communication is experienced during actual exercise

situations. For example, researchers could observe participants during workouts, runs, or recovery routines and then discuss emotionally significant moments immediately afterwards. Such approaches could reduce reliance on retrospective accounts while still preserving users' own interpretations of their experiences.

Future research can build on this study by examining its interpretive findings across broader samples, longer timeframes, narrower app contexts, and increasingly AI-mediated health, fitness, and wellbeing environments. This study offers a communication-centred foundation for understanding how emotionally significant app experiences are shaped, but further research is needed to examine how these dynamics develop across different users, technologies, and stages of long-term app use.

To conclude the thesis, this study provides a communication-centred interpretation of emotionally significant app experiences in health and exercise applications. While the findings are limited to a small qualitative sample, they show how app-based communication may shape users' motivation, trust, frustration, validation, and sense of agency through both explicit and implicit communicative cues. The study therefore suggests that emotionally intelligent app communication involves more than adding encouragement or personalisation. It requires communication that is context-sensitive, intelligible, respectful, and appropriately restrained. In this sense, emotionally engaging user journeys are shaped not only by what applications say to users, but also by how they structure interaction, explain their logic, respond to bodily experience, and leave room for users' own interpretation.

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APPENDICES

Appendix 1. Use of artificial intelligence in the research process

This thesis was prepared with the support of generative AI and AI-enhanced tools. These tools were used between autumn 2024 and spring 2026 as supplementary aids for language refinement, literature discovery, structural development, analytical reflection, consistency checking, and quality review. The purpose of using these tools was to support, rather than replace, the author's academic reasoning, methodological decision-making, analysis, and responsibility for the final thesis.

All AI-generated suggestions were critically evaluated by the author. The tools were not used to conduct autonomous analysis, make final methodological decisions, generate final themes independently, or replace the author's interpretation of the empirical data. Final decisions concerning the research design, incident selection, coding refinement, theme development, interpretation of findings, selection of quotations, and final written argument were made by the author.

TABLE A1. Use of artificial intelligence in the research process.

Used tool	Purpose of use	Examples of prompts or tasks
ChatGPT by OpenAI	Used throughout the thesis process as a supplementary tool for academic language refinement, paragraph restructuring, argument clarification, reduction of repetition, and critical feedback on thesis sections. Also used to support methodological reflection, development of interview and analysis materials, consistency checking, and final quality review. During the analysis phase, ChatGPT was used to support the organisation and critical review of researcher-produced analytical material, such as coding notes, analysis workbooks, researcher memos, and emerging theme structures.	Examples of tasks included: "Improve the academic clarity of the following paragraph while preserving the meaning"; "Review this thesis section for coherence, repetition, and alignment with the research questions"; "Identify possible weaknesses in this methodological explanation"; "Suggest clearer wording for this theme definition"; "Compare these preliminary themes and identify possible overlap or inconsistency."

Used tool	Purpose of use	Examples of prompts or tasks
Gemini by Google	Used as an additional AI tool for comparison, critical review, and alternative feedback on selected thesis sections and analytical interpretations. Gemini was used to obtain a second perspective on structure, clarity, argumentation, and thematic coherence. Its outputs were compared with other tools and with the original data.	Examples of tasks included: "Review this chapter and identify the most important remaining coherence issues"; "Give concise feedback on whether the research questions, theory, methodology, findings, and discussion align"; "Identify possible blind spots in this interpretation."
NotebookLM Pro by Google	Used to support the organisation and review of uploaded thesis-related materials, such as thesis drafts, methodological notes, feedback documents, and selected source materials. NotebookLM was used to summarise, compare, and navigate materials, not to replace the author's reading or interpretation.	Examples of tasks included: "Summarise the main points of the uploaded materials"; "Identify where this uploaded document discusses methodological trustworthiness"; "Compare these thesis sections and identify possible inconsistencies."

Used tool	Purpose of use	Examples of prompts or tasks
Keenious	Used during the literature review process to identify potentially relevant academic sources and refine literature search directions. Sources suggested through Keenious were evaluated by the author before inclusion in the thesis. Keenious was used as a literature discovery aid, not as a substitute for source evaluation.	Examples of tasks included searching for literature related to customer journey, emotional journey, emotional intelligence, critical incidents, app-based communication, digital service engagement, and health and fitness applications.
Thesify AI	Used to support academic quality review, language refinement, and thesis-level feedback on selected sections. The tool was used to identify possible issues in clarity, structure, argumentation, academic tone, and consistency.	Examples of tasks included requesting feedback on thesis structure, academic clarity, methodological coherence, and possible areas for improvement in selected thesis sections.

Used tool	Purpose of use	Examples of prompts or tasks
University of Jyväskylä MyJYU AI transcription tool	Used to support the initial transcription of Finnish-language interview data. The transcripts were checked and corrected by the author before analysis.	Used for initial transcription of recorded interviews. The resulting transcripts were reviewed against the original interview recordings and corrected where necessary.

External, non-university AI tools were used only after empirical data had been anonymised to remove personally identifiable information. AI-generated outputs were treated as provisional suggestions for reflection, comparison, and language development. Suggestions that were unsupported, overly general, inconsistent with the data, or insufficiently aligned with the research questions and corporate communication focus of the thesis were rejected or revised.

During the thematic analysis process, AI tools were used only as supplementary aids for organising, comparing, questioning, and refining researcher-produced analytical material. They were not used to conduct independent coding, determine the final themes, select quotations autonomously, or make final analytical claims. All interpretations were checked against the original Finnish transcripts, the structured analysis workbook, researcher memos, the research questions, and relevant methodological literature.

Selected quotations used in the findings chapter were verified against the original Finnish transcripts and translated into English by the author. The author remained responsible for the interpretation of the data, the accuracy of the final text, the integrity of the argument, and all scholarly decisions made in the thesis.

Appendix 2. The interview guide

Study: Examining the emotional journey of health and exercise application users
Interview type: Semi-structured stimulated recall interview with app walkthrough, optional Plutchik's Wheel of Emotions, and Critical Incident Technique
Target duration: approximately 45 minutes; extended interviews up to approximately 60 minutes
Interview language: Finnish

The interview guide below was used flexibly. Not all sub-questions were asked in every interview, as participants often provided data-rich answers through the main questions and follow-up discussion. The Critical Incident Technique section was repeated for each selected incident. The aim was to examine at least three emotionally significant incidents per participant, with additional incidents discussed when time and participant willingness allowed.

1. Opening and ethical briefing

Q1.1 Käydäänkö nopeasti läpi tiedote, suostumus ja äänitys – ja varmistetaan, että tämä on sinulle ok?

- Voit keskeyttää milloin tahansa tai ohittaa kysymyksiä.
- Näytät sovellusta vain siltä osin kuin itse haluat.
- Haastattelu tehdään suomeksi; gradu kirjoitetaan englanniksi ja mahdolliset lainaukset käännetään.

Q1.2 Onko sinulla ennen aloitusta kysymyksiä tutkimuksesta tai tietosuojasta?

2. Background information

Interviewer note: pidä tiiviinä. Tämä on taustakonteksti, ei keskustelun ydin.

Q2.1 Mitä sovellusta käytät tässä treenissä?

- Jos et halua sanoa nimeä, voit kuvata vain sovelluksen tyypin, esimerkiksi juoksu-, sali- tai yleistracker-sovellus.

Q2.2 Kuinka kauan olet käyttänyt sitä?

- alle 3 kk / 3–12 kk / yli 1 v

Q2.3 Kuinka usein käytät sitä viikossa?

- 1–2 / 3–4 / 5+ / vaihtelee

Q2.4 Onko sovelluksen push-ilmoitukset yleensä päällä vai pois?

- Minkä tyyppisiä push-ilmoituksia yleensä tulee?

Q2.5 Mihin ikäryhmään kuulut?

- 18–20 / 21–23 / 24–26 / 27–30 / 31+

Q2.6 Mikä sukupuoli kuvaa sinua parhaiten? Tämä kysymys on vapaaehtoinen.

- nainen / mies / muu / en halua vastata

Q2.7 Mikä rooli tällä sovelluksella on sinulle?

- tracker / valmentaja / motivaattori / rutiinityökalu / yhteisö / muu

3. Context of the selected exercise experience

Q3.1 Kerro lyhyesti, millainen treeni oli ja miten sovellus liittyi siihen.

- Oliko tämä tavallinen vai poikkeava treeni?
- Käytitkö sovellusta ennen treeniä, sen aikana, sen jälkeen vai vain yhdessä vaiheessa?

Q3.2 Mitä toivoit tältä treeniltä?

- suoritus / palautuminen / rutiini / stressin purku / "vain tehdä jotain"

Q3.3 Millä fiiliksellä aloitit treenin ja avasit sovelluksen?

- Esimerkiksi innostunut, neutraali, väsynyt, paineinen, epävarma, utelias.
- Muuttuiko fiilis treenin aikana tai jälkeen?

Q3.4 Avasitko sovelluksen oma-aloitteisesti vai jonkin push-ilmoituksen takia?

- Jos ilmoitus vaikutti avaamiseen, mitä se suunnilleen sanoi ja milloin se tuli?

4. Optional Plutchik emotion wheel support

Q4.1 Käytän haastattelussa apuna tunnepyörää. Haluatko käyttää sitä tunteiden nimeämiseen vai käytätkö mieluummin omia sanoja?

- Pyörä on vapaaehtoinen: voit käyttää omia sanoja, ja pyörä on tukena, jos haluat tarkentaa.
- Voimakkuutta voidaan kuvata esimerkiksi sanoilla lievä, keskitaso tai voimakas, tai halutessasi asteikolla 1–5.

5. App walkthrough: exercise journey

Interviewer note: virikehaastattelun ankkuri. Pysähdy vain, jos jokin kohta herättää tunteita tai reaktioita. Kysy tunne ja voimakkuus silloin, kun osallistuja ilmaisee selkeän reaktion tai muutoskohdan.

Q5.0 Käydään treeni läpi vaihe vaiheelta sovelluksessa: näytä mistä näet treenin tai yhteenvedon ja etene omin sanoin. Minä kysyn väliin tarkentavia kysymyksiä.

Trigger and entry
Q5.1 Mikä oli viimeinen sysäys juuri ennen kuin avasit sovelluksen?

- Oliko kyseessä push-ilmoitus, rutiini, tarve vai jokin muu syy?
- Mitä ajattelit sovelluksesta sillä hetkellä?

Set-up
Q5.2 Mitä teit sovelluksessa ennen kuin treeni alkoi?

- Valitsitko treenin, painoitko start, valitsitko tavoitteen tai tarkistitko jotain?
- Tarvittaessa: mikä tunne tästä syntyi ja kuinka voimakas se oli?

In-session interaction
Q5.3 Mitä sovellus "teki" treenin aikana?

- Antoiko se ohjeita, muistutuksia, mittareita, keskeytyksiä tai kitkaa?
- Katsoitko ruutua usein vai harvoin?
- Tarvittaessa: mikä tunne tästä syntyi ja kuinka voimakas se oli?

Immediate post-session feedback
Q5.4 Mitä sovellus näytti tai sanoi heti treenin jälkeen?

- Palaute, yhteenveto, saavutukset, streak, badge tai muu palaute?
- Tarvittaessa: mikä tunne tästä syntyi ja kuinka voimakas se oli?

Reflection
Q5.5 Mitä tämä treeni ja sovelluksen palaute "tarkoitti sinulle"?

- Tuntuiko se onnistumiselta, epäonnistumiselta vai ei juuri miltään?
- Tarvittaessa: mikä tunne tästä syntyi ja kuinka voimakas se oli?

Social interaction
Q5.6 Tuliko tähän treeniin liittyen mitään jakamista, vertailua tai reaktioita?

- Jos ei, tämä kohta voidaan ohittaa.
- Tarvittaessa: mikä tunne tästä syntyi ja kuinka voimakas se oli?

Next-step nudges
Q5.7 Ehdottiko sovellus jotain seuraavaksi tai tuliko myöhemmin ilmoituksia?

- Esimerkiksi muistutus, seuraavan päivän ehdotus, streak-varoitus tai suositus.
- Tarvittaessa: mikä tunne tästä syntyi ja kuinka voimakas se oli?

Exit and aftertaste
Q5.8 Millainen fiilis jäi, kun lopetit sovelluksen käytön?

- Vetikö sovellus takaisin vai työnsikö se pois?

6. Selection of critical incidents

Interviewer note: kirjaa incidentit I1–I5. Tavoitteena on valita ensin kolme incidenttiä. Lisäincidentit käsitellään vain, jos aikaa on.

Q6.1 Valitaan tästä treenistä kolme hetkeä, jotka olivat tunteellisesti selkeitä tai merkityksellisiä. Mitkä ne ovat?

- paras tai kannustavin hetki
- ärsyttävin tai kuormittavin hetki
- hetki, jossa motivaatio muuttui ylös tai alas, tai joka tuntui ristiriitaiselta
- Incident voi olla myös kohta, jossa sovellus ei viestinyt mitään, vaikka odotit ohjausta tai palautetta.

Q6.2 Valitaanko vielä yksi tai kaksi hetkeä lisää?

- Vain jos aikaa on.
- Esimerkiksi yllätyshetki tai hetki, joka kuvaa parhaiten koko sovelluksen vaikutusta.

Incident log used by the interviewer

	Phase	**Communicative element**	**Emotion**	**Valence**	**Intensity**	**Short title**
I1						
I2						
I3						
I4						
I5						

7. CIT deep-dive questions

The following questions were repeated for each selected incident. The minimum aim for each incident was to cover:

1. the concrete moment, view, or message;
2. the communicative element or its absence;
3. interpretation, emotion, and intensity;
4. behavioural consequence; and
5. one improvement suggestion.

Event and anchoring
Q7.1 Kerro mitä tässä hetkessä tapahtui – askel askeleelta.

- Missä kohtaa treeniä oltiin?
- Voitko näyttää sovelluksesta sen kohdan tai näkymän?

Communicative element
Q7.2 Mitä sovellus tässä hetkessä viesti tai "teki" sinulle?

- Oliko tämä push-ilmoitus vai sovelluksen sisäinen palaute?
- Oliko se ohjaus tai ehdotus, tulos tai yhteenveto, palkinto tai streak, vertailu, personointi vai kitka tai virhe?

Format, timing, and tone
Q7.3 Miten se tuli ja milloin se tuli?

- tekstinä / äänenä / värinä / visuaalisena / numerodatan kautta
- oikeaan aikaan / keskeyttävästi / liian myöhään / liian aikaisin
- Millainen sävy siinä oli?
- neutraali / kannustava / ohjaava / arvioiva / varoittava / humoristinen / "streak-henkinen"

Interpretation
Q7.4 Mitä se viesti sinulle – miten sinä sen tulkitsit?

- Mitä se "yritti saada sinut tekemään"?
- Mitä se mahdollisesti "kertoi sinusta"?
- Ymmärsinkö oikein, että…?

Interviewer note: tiivistä osallistujan vastaus ja pyydä korjaamaan tarvittaessa.

Emotions
Q7.5 Mikä tunne sinussa heräsi tässä hetkessä?

- Oma sana ensin; halutessasi Plutchik.
- Kuinka voimakas se oli? Esimerkiksi lievä, keskitaso tai voimakas, tai halutessasi asteikolla 1–5.
- Oliko kyseessä yksi tunne vai tunteiden sekoitus?

Behavioural consequence
Q7.6 Mitä teit tämän takia?

- Mitä teit heti seuraavien minuuttien aikana?
- Jatkoitko, lopetitko, keskeytitkö, ohititko tai selvititkö asiaa?
- Mitä teit seuraavina päivinä tai seuraavalla treenikerralla?
- Palasitko, lykkäsitkö, vältitkö tai muuttiko se rutiinia?
- Jos tällaisia hetkiä toistuisi, miten se vaikuttaisi pitkässä juoksussa motivaatioon ja sovelluksen käyttöön?

Context
Q7.7 Miksi luulet, että tämä tuntui juuri silloin tuolta?

- Esimerkiksi väsymys, stressi, tavoite, aiemmat kokemukset, treenin vaikeus, itseluottamus, ulkoinen paine tai päivän mieliala.
- Olisiko se tuntunut erilaiselta toisena päivänä?

Improvement suggestion
Q7.8 Miten sovelluksen olisi pitänyt viestiä tässä tilanteessa, jotta se olisi ollut sinulle toimivampi tai hyödyllisempi?

- sanamuoto / ajoitus / sävy / määrä / personointi / palaute vs. ilmoitus
- Mikä olisi todennäköisemmin saanut sinut jatkamaan tai avaamaan sovelluksen uudelleen?

Communicative element type, interviewer note

The interviewer used the following categories as a note-taking aid:

- Push-ilmoitus
- In-app palaute
- Ohjaus tai ehdotus
- Tulokset tai yhteenveto
- Palkinto, streak, badge tai pisteet
- Vertailu: itse, aiemmat suoritukset, muut käyttäjät tai tavoite
- Personointi
- Kitka, virhe tai epäselvyys
- Muu

8. Final synthesis

Q8.1 Mitkä sovelluksen viestit tai tilanteet yleensä lisäävät motivaatiota ja mitkä vähentävät?

- Esimerkki kummastakin.

Q8.2 Missä vaiheessa polkua sovellus on sinulle parhaimmillaan ja missä heikoimmillaan?

- Avaus, treenin aikainen käyttö, palaute, seuraava askel, ilmoitukset tai muu vaihe.

Q8.3 Onko jotain tärkeää, jota en kysynyt, mutta joka vaikuttaa siihen, miltä tämä sovellus tuntuu sinusta?

9. Debrief and pilot feedback questions

Q9.1 Haluatko, että jokin kohta jätetään pois aineistosta?
Q9.2 Tuntuiko jokin kysymys epäselvältä tai johdattelevalta?
Q9.3 Oliko haastattelun pituus sinulle ok?
Q9.4 Onko sinulla vielä kysymyksiä tutkimuksesta?